

Long–Short Portfolio Return Book

Cumulative return of each factor's long–short
portfolio, net of 20 bp transaction cost.
225 factors over 104 months, ordered by CAGR.

HOW TO READ

Each curve starts at 0 % in December 2017 and
compounds monthly long–short returns. The line is
coloured by the factor's style; the figure at the
right edge is the period-end cumulative return.

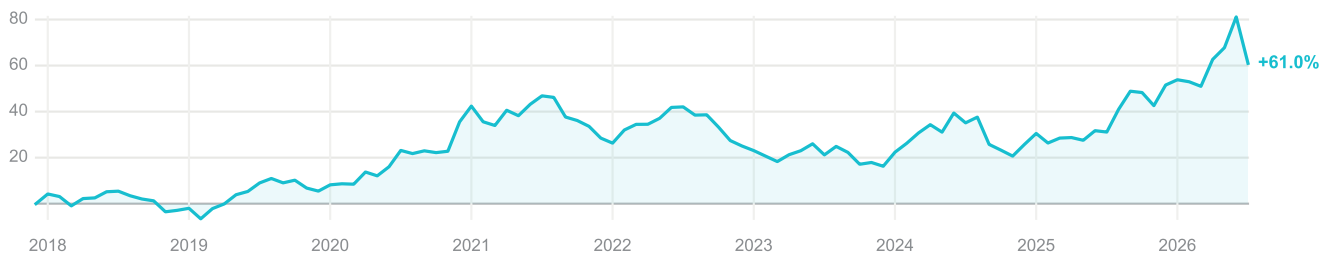
FACTOR STYLES

● Valuation	35	● Price Momentum	50
● Earnings Quality	46	● Size	4
● Analyst Expectations	18	● Historical Growth	38
● Capital Efficiency	27	● Volatility	7

001 ● **3M Revision in FY2 EPS Estimate**

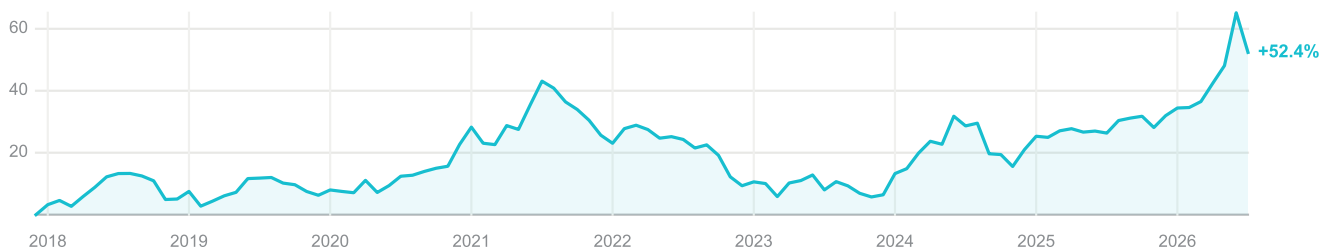
Analyst Expectations

CAGR +7.4%

002 ● **Street Revision Magnitude**

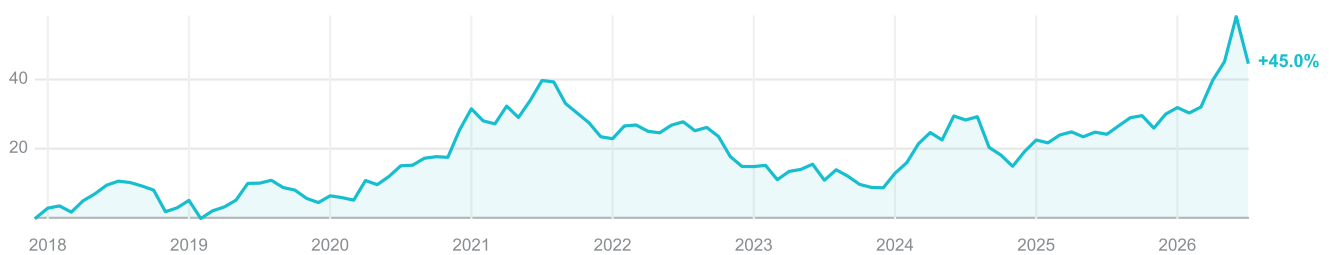
Analyst Expectations

CAGR +7.1%

003 ● **3M Revision in FY1 EPS Estimate**

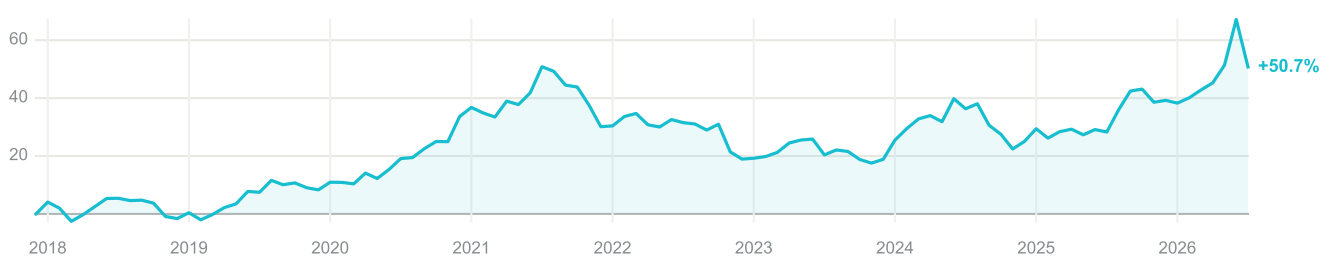
Analyst Expectations

CAGR +6.9%

004 ● **Number of EPS FY1 Revisions**

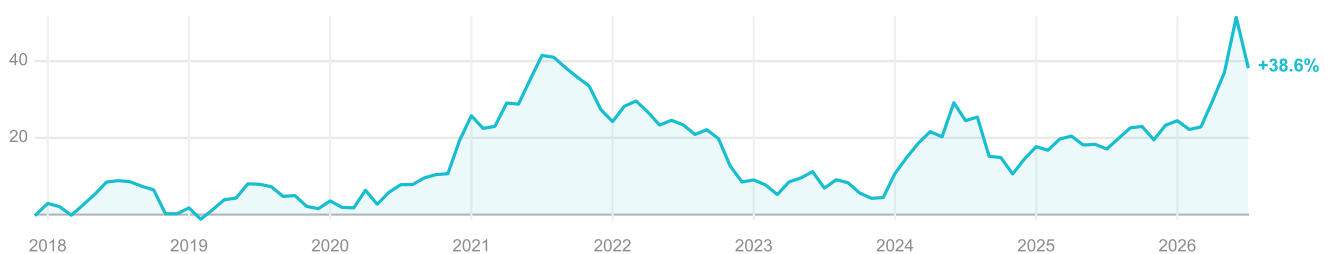
Analyst Expectations

CAGR +6.6%

005 ● **Adjusted Revision Magnitude**

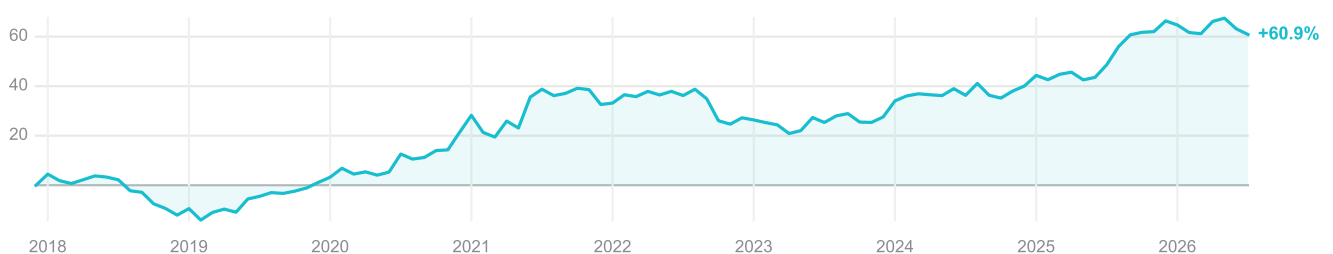
Analyst Expectations

CAGR +6.2%

006 ● **Standardized Unexpected Earnings**

Analyst Expectations

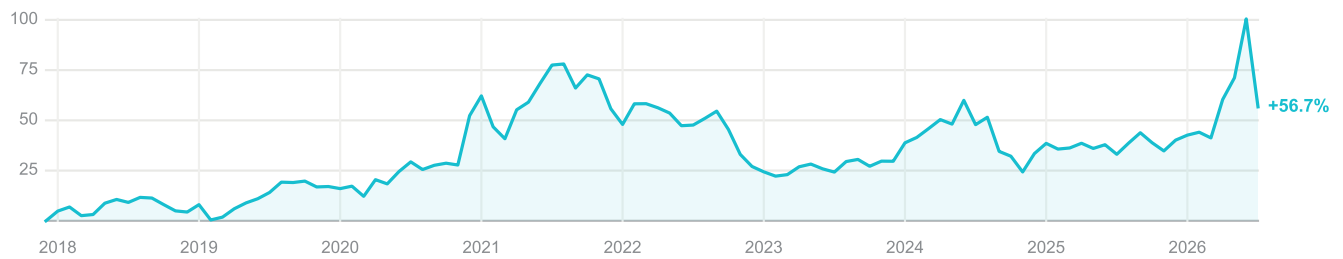
CAGR +5.9%



007 ● 6M Chg in Target Price

Analyst Expectations

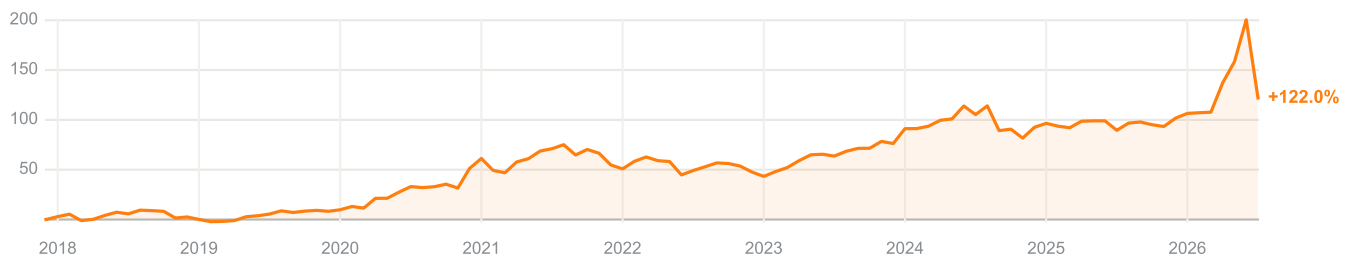
CAGR +5.8%



008 ● Slope of 52W Price Trend

Price Momentum

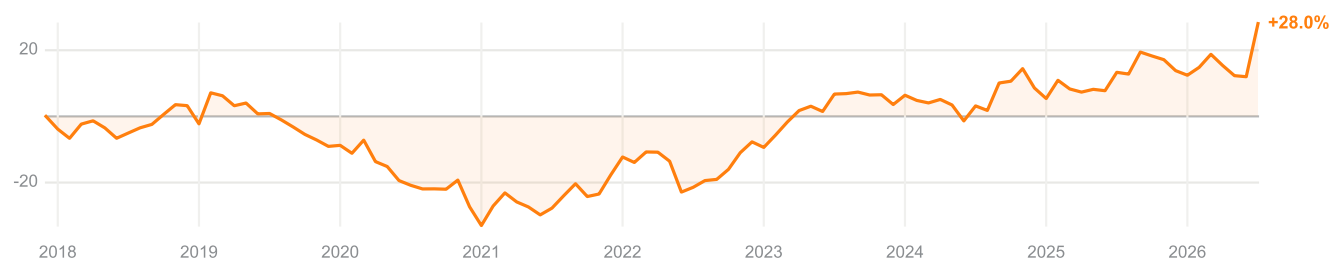
CAGR +5.7%



009 ● Sharpe Ratio

Price Momentum

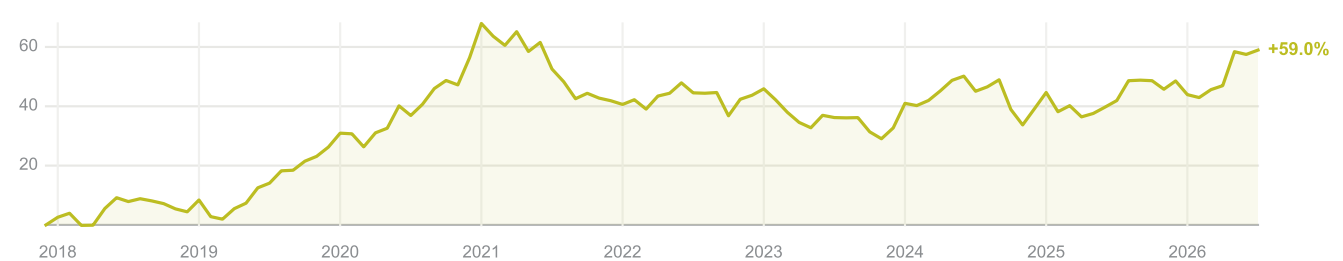
CAGR +5.3%



010 ● Cash Flow Return on Invested Capital

Capital Efficiency

CAGR +5.2%



011 ● Operating Cash Flow to Enterprise Value

Valuation

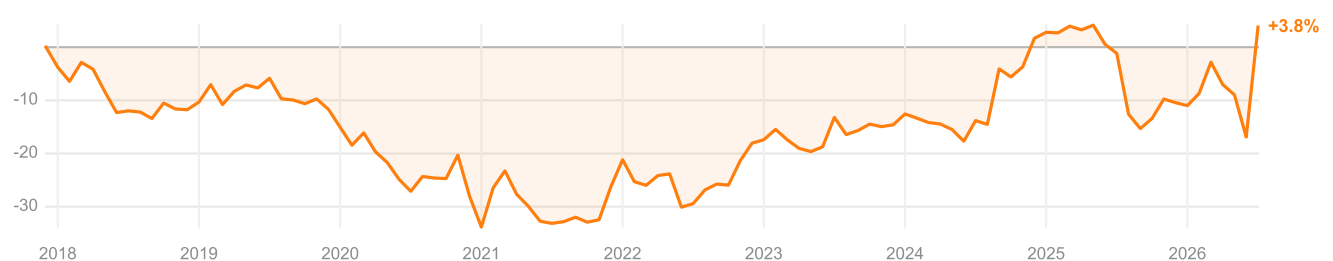
CAGR +5.0%



012 ● Closing Price to 260 Day Low

Price Momentum

CAGR +5.0%



013 ● Dividend per Share Growth

Historical Growth

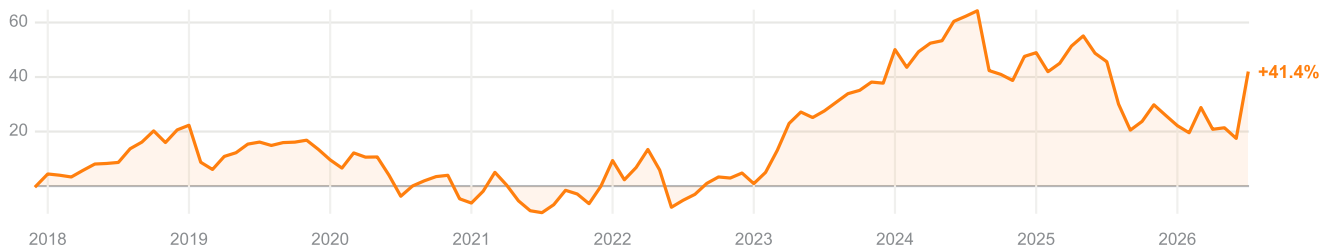
CAGR +5.0%



014 ● 24M Residual Return Variance

Price Momentum

CAGR +5.0%



015 ● 3Y Chg in Earnings per Share

Historical Growth

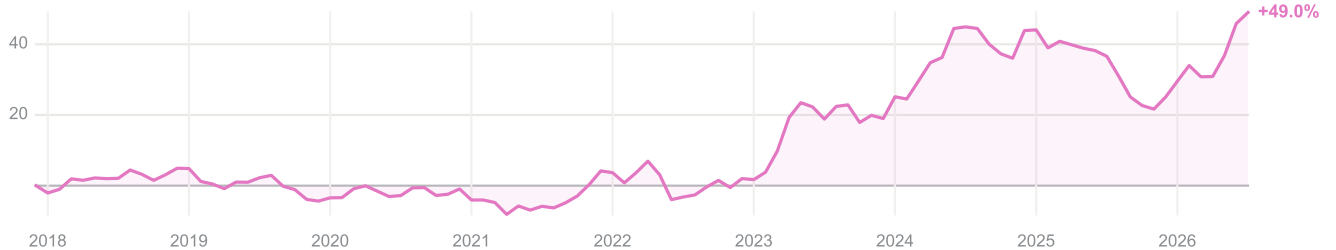
CAGR +4.8%



016 ● ROE 20 Qtr Standard Deviation

Earnings Quality

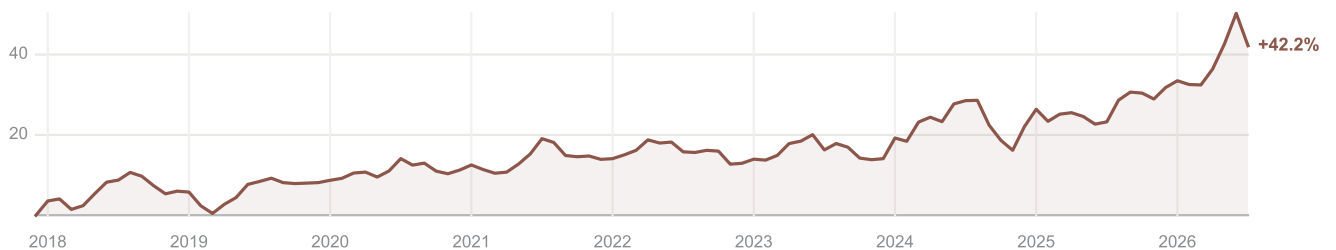
CAGR +4.8%



017 ● 1Y Chg in Sales Turnover

Historical Growth

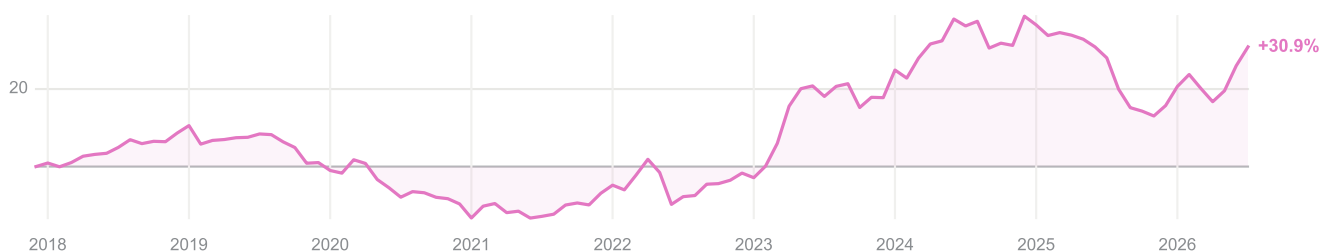
CAGR +4.7%



018 ● ROA 20 Qtr Standard Deviation

Earnings Quality

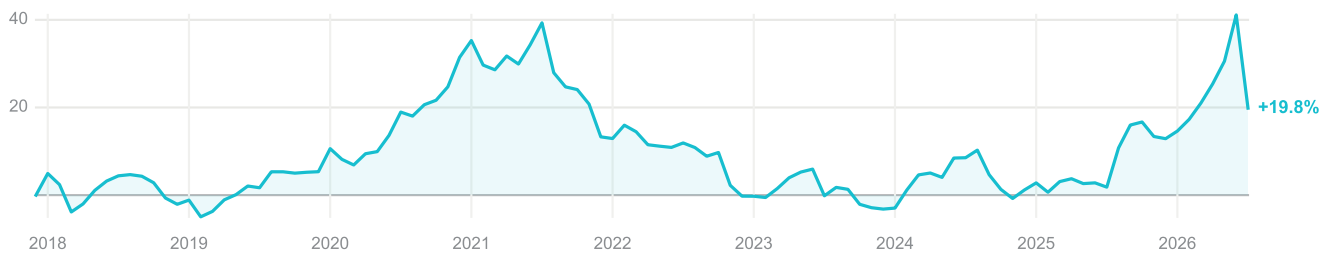
CAGR +4.5%



019 ● Adjusted Number of EPS FY1 Revisions

Analyst Expectations

CAGR +4.5%



020 ● Operating Cash Flow to Equity

Capital Efficiency

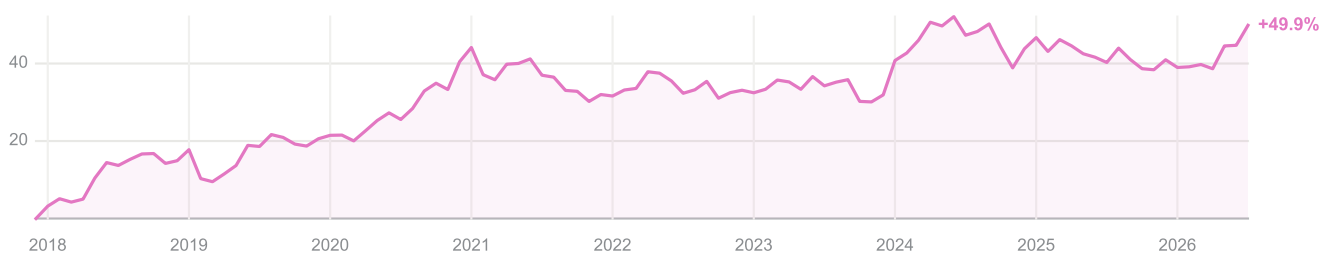
CAGR +4.4%



021 ● Free Cash Flow to Sales

Earnings Quality

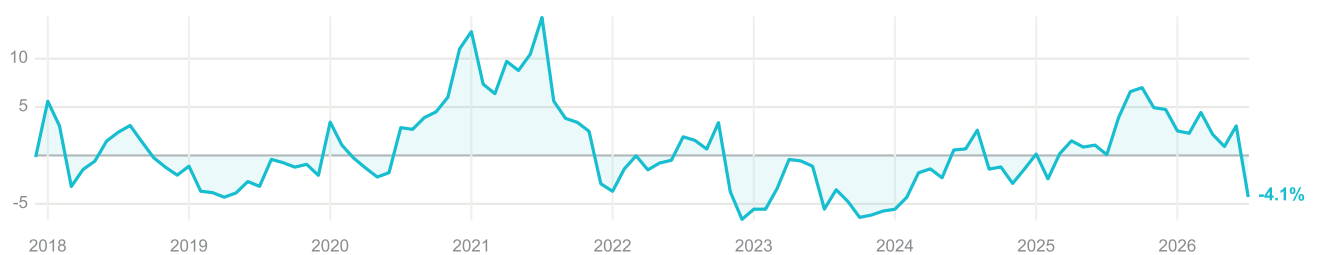
CAGR +4.3%



022 ● Analyst Earnings Estimate Diffusion

Analyst Expectations

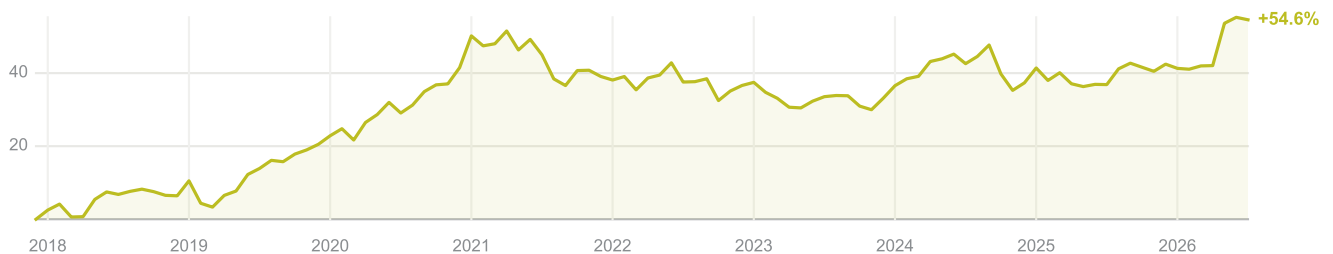
CAGR +4.3%



023 ● Operating Cash Flow Ratio

Capital Efficiency

CAGR +4.3%



024 ● Sales to Price Ratio

Valuation

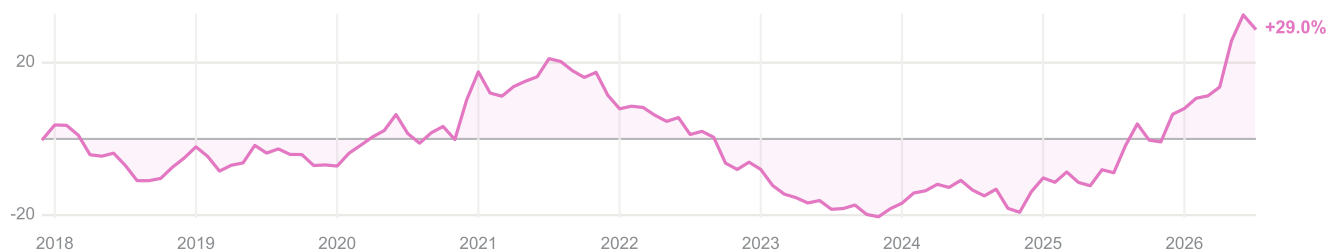
CAGR +4.3%



025 ● Net Profit Margin

Earnings Quality

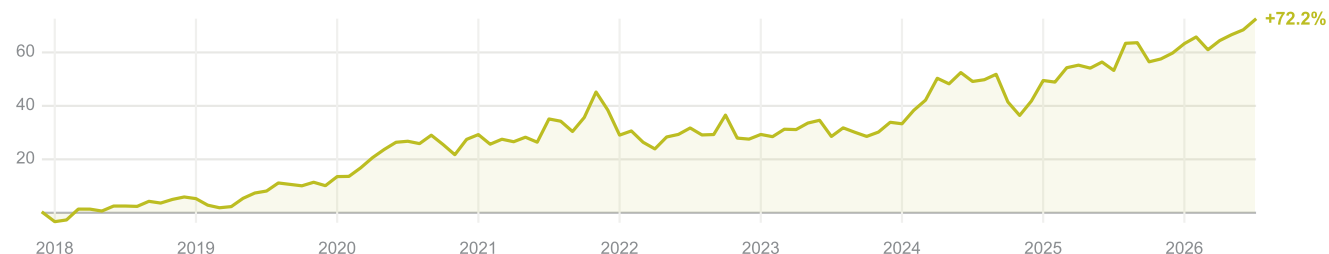
CAGR +4.2%



026 ● Long Term Debt to Equity Ratio

Capital Efficiency

CAGR +4.2%



027 ● 1Y Chg in Operating Cash Flow per Share

Historical Growth

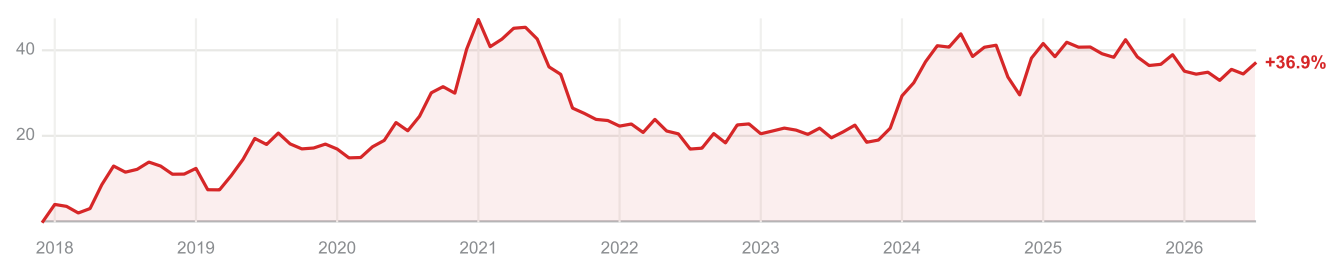
CAGR +4.2%



028 ● Free Cash Flow to Common Equity

Valuation

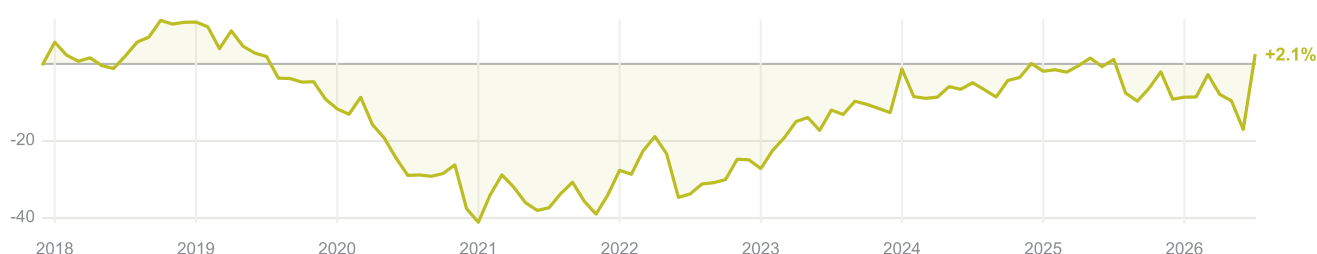
CAGR +3.9%



029 ● Tobin's Q Ratio

Capital Efficiency

CAGR +3.9%



030 ● Operating Cash Flow to Price

Valuation

CAGR +3.8%



031 ● **Current Liability to Price**

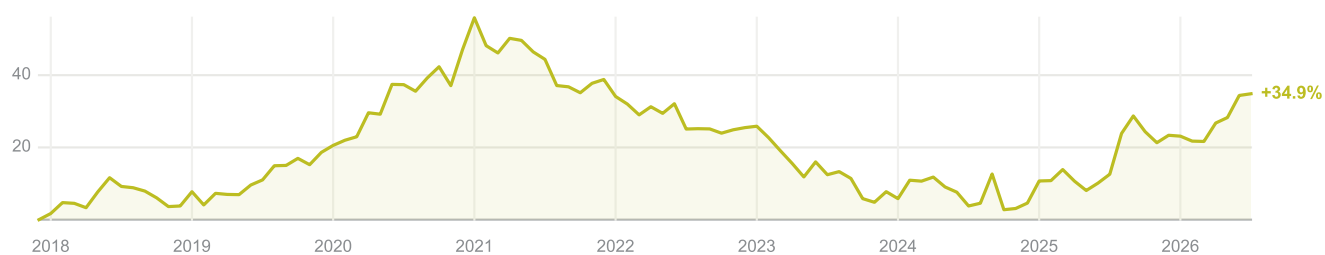
Valuation

CAGR +3.8%

032 ● **Gross Profit to Assets**

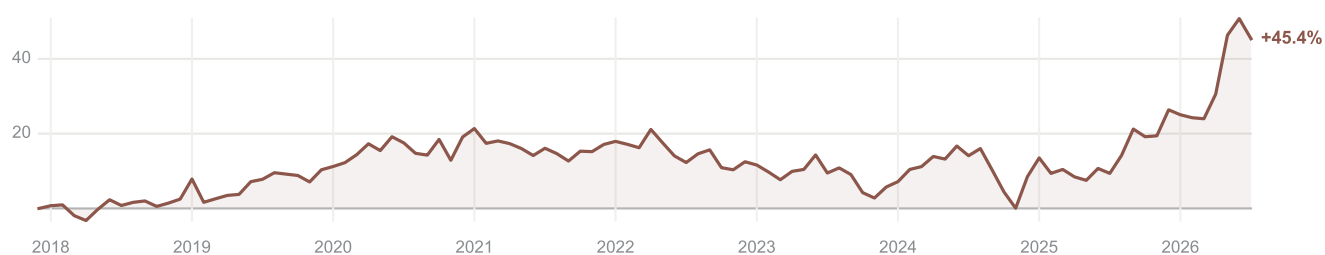
Capital Efficiency

CAGR +3.7%

033 ● **3Y Chg in Operating Margin**

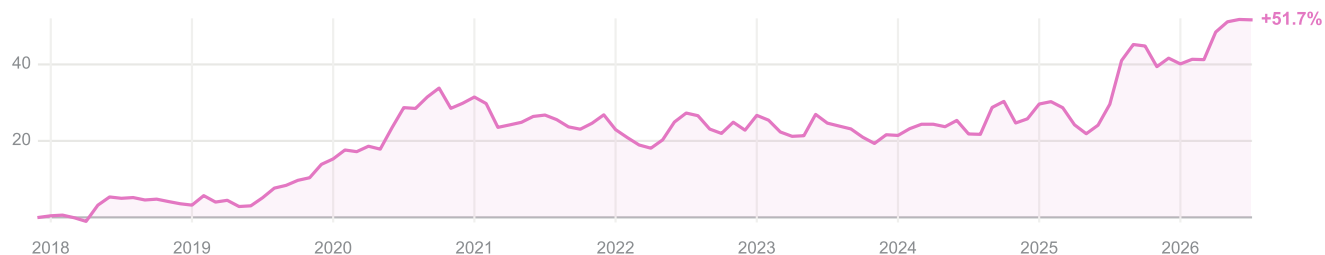
Historical Growth

CAGR +3.7%

034 ● **Asset Turnover**

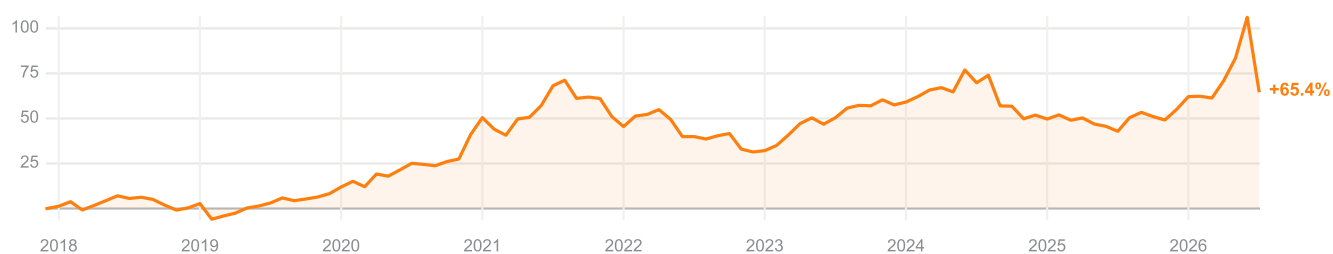
Earnings Quality

CAGR +3.7%

035 ● **180D Price TStat**

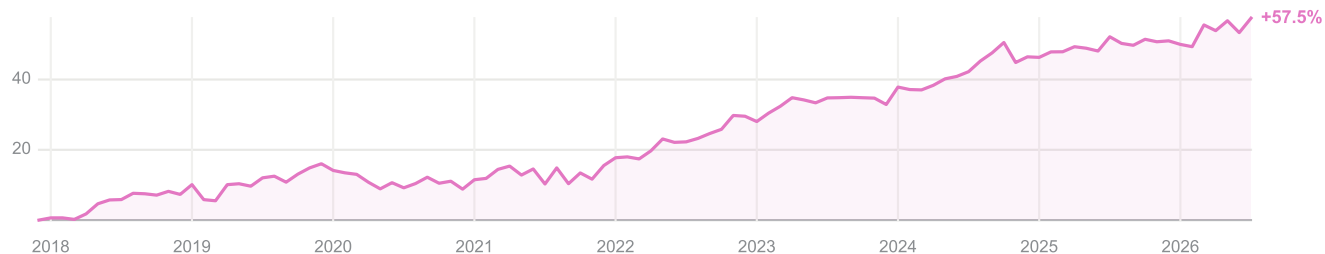
Price Momentum

CAGR +3.7%

036 ● **Adjusted Accruals**

Earnings Quality

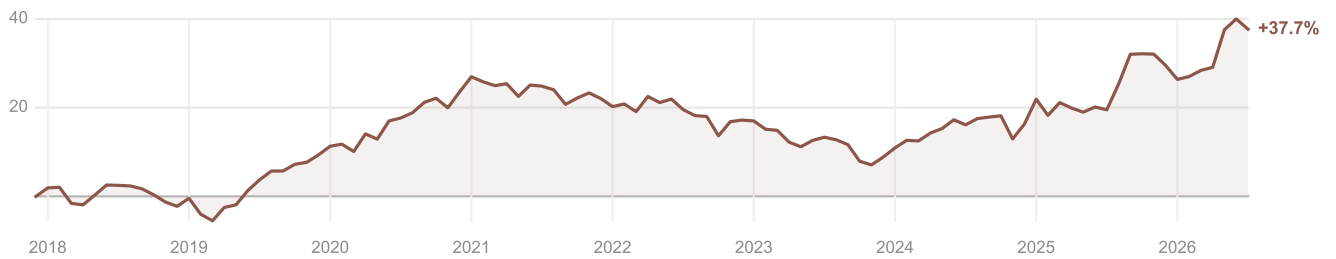
CAGR +3.7%



037 ● 3Y Chg in Operating Cash Flow per Share

Historical Growth

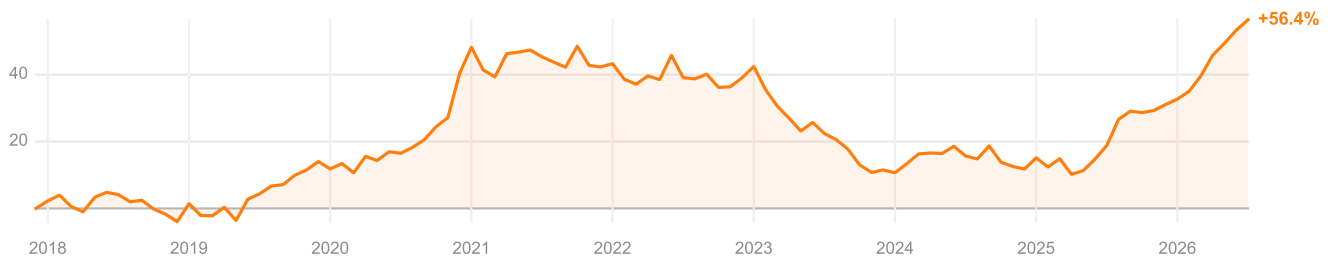
CAGR +3.6%



038 ● Institution Ownership Concentration

Price Momentum

CAGR +3.6%



039 ● Target Price Gap to 6M EMA

Analyst Expectations

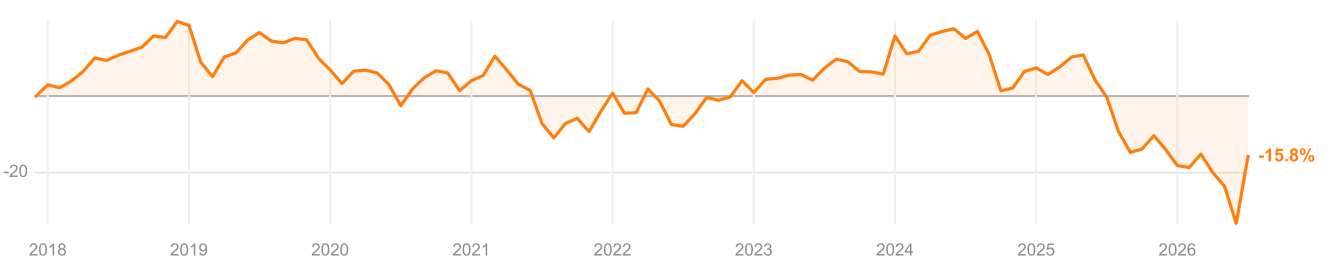
CAGR +3.5%



040 ● Share Turnover

Price Momentum

CAGR +3.5%



041 ● 1Y Chg in Sales to Price

Historical Growth

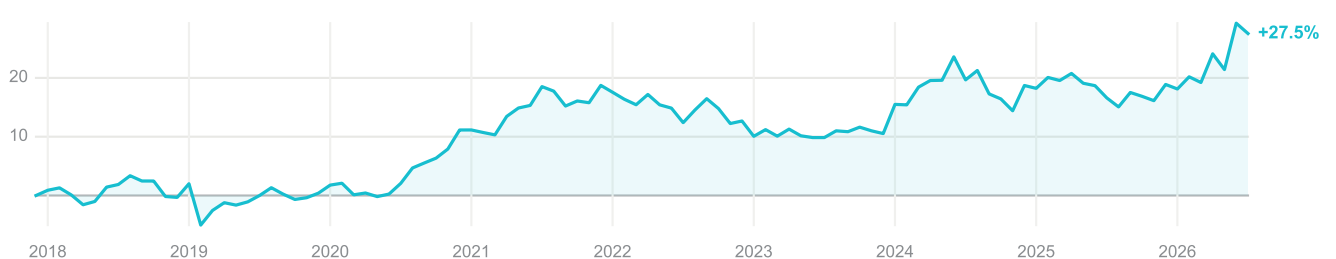
CAGR +3.5%



042 ● 6M Avg Chg 1M Recommendation

Analyst Expectations

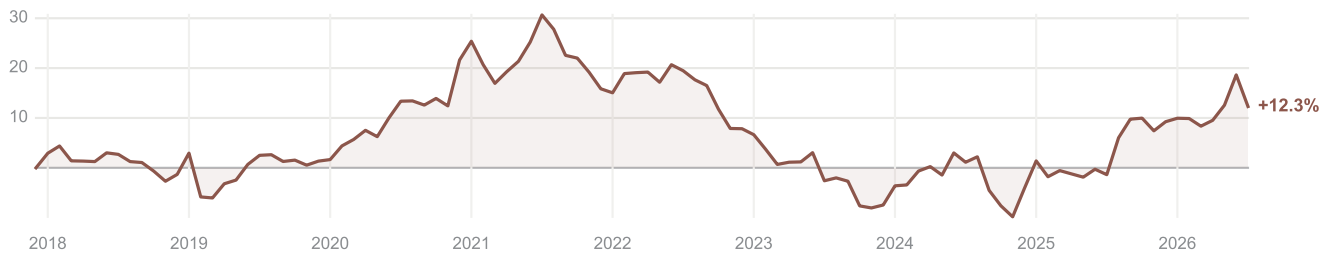
CAGR +3.4%



043 ● 1Y Chg in Asset Adjusted EPS

Historical Growth

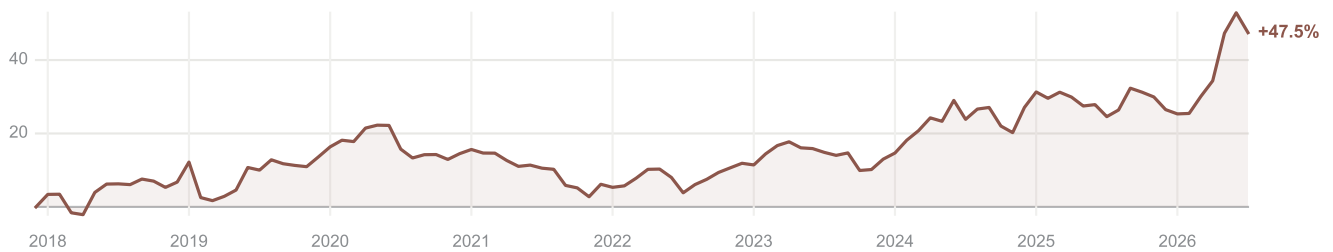
CAGR +3.3%



044 ● 3Y Chg in Free Cash Flow per Share

Historical Growth

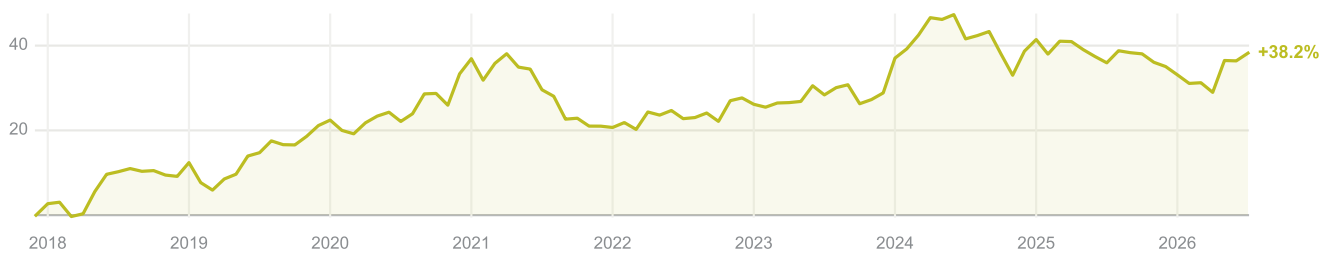
CAGR +3.3%



045 ● Capital Acquisition Ratio

Capital Efficiency

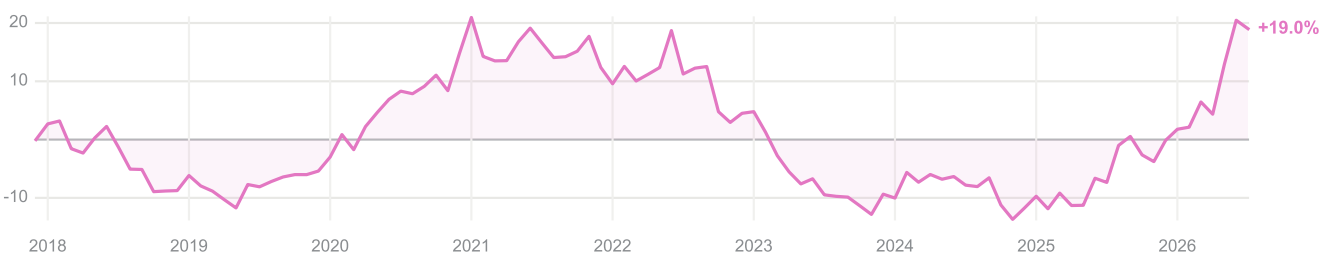
CAGR +3.2%



046 ● EBIT Margin

Earnings Quality

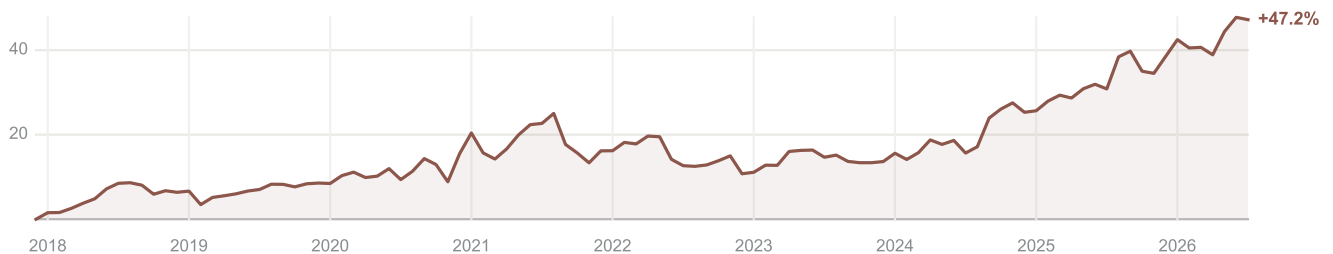
CAGR +3.1%



047 ● 1Y Chg in Asset Adjusted Free Cash Flow

Historical Growth

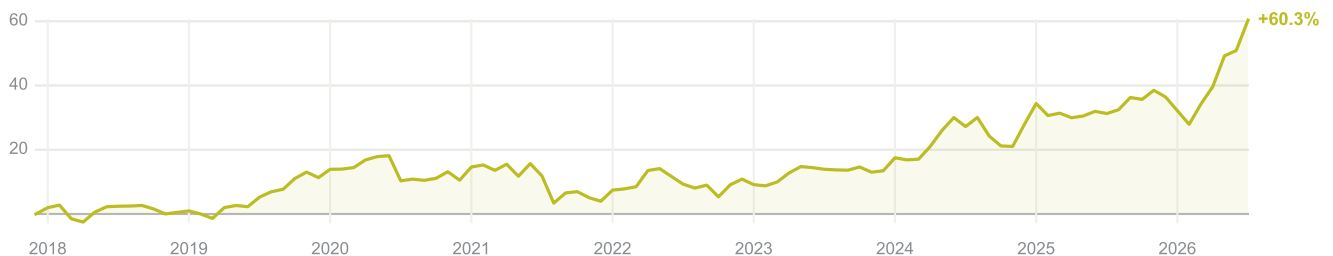
CAGR +3.1%



048 ● Receivables Turnover Ratio

Capital Efficiency

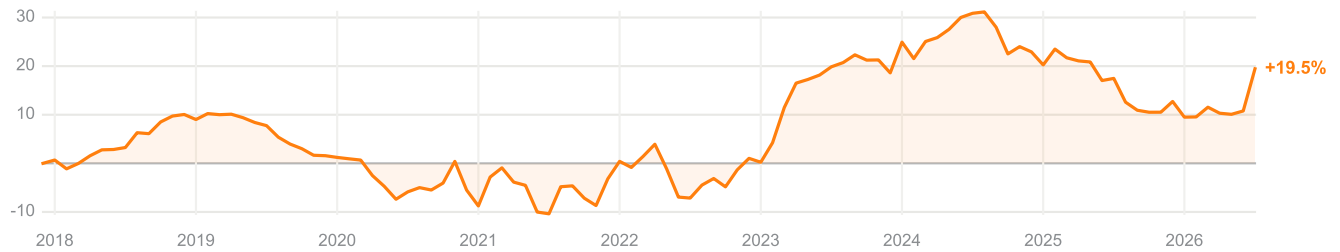
CAGR +3.1%



049 ● **Ownership Breadth Stability of Institution Ownership by Holders**

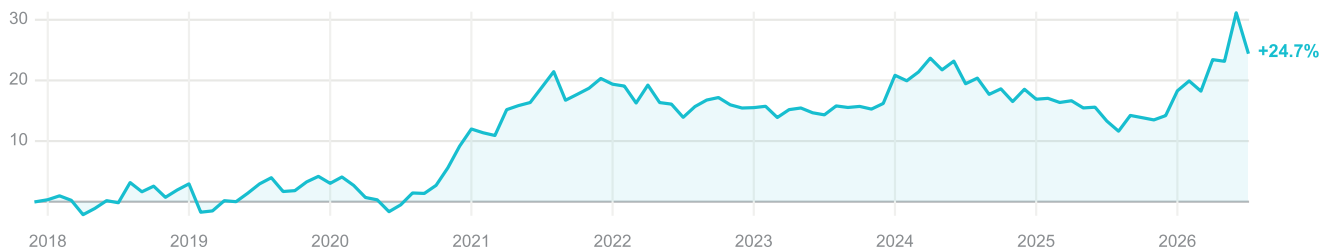
Price Momentum

CAGR +3.1%

050 ● **Buy-to-Sell Recommendation Ratio Less 3M EMA**

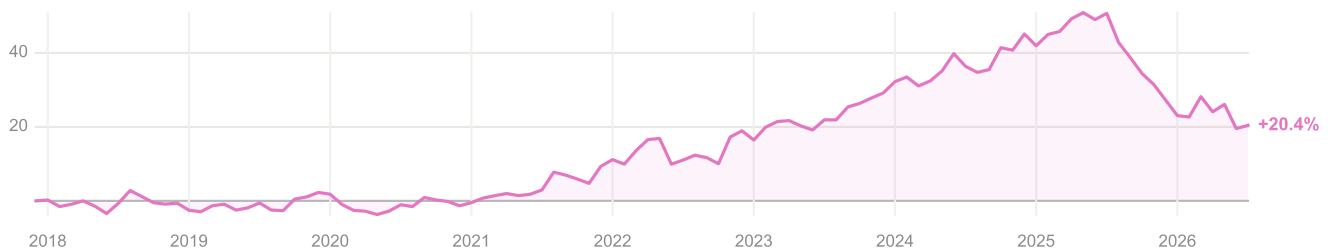
Analyst Expectations

CAGR +3.1%

051 ● **Working Capital Accruals**

Earnings Quality

CAGR +3.1%

052 ● **Assets to Price Ratio**

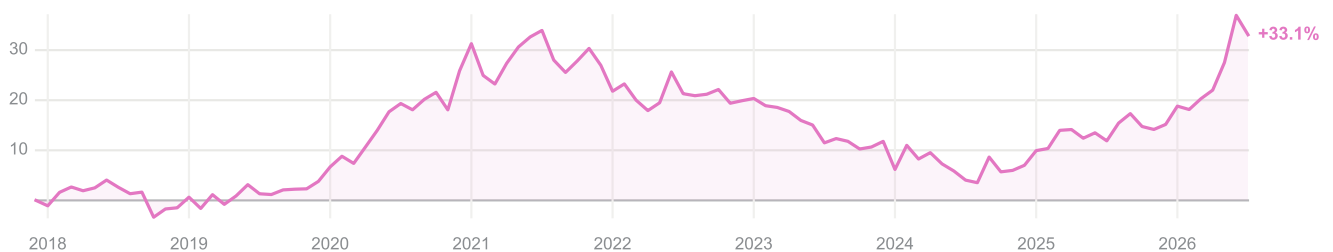
Valuation

CAGR +3.0%

053 ● **Gross Profit Margin**

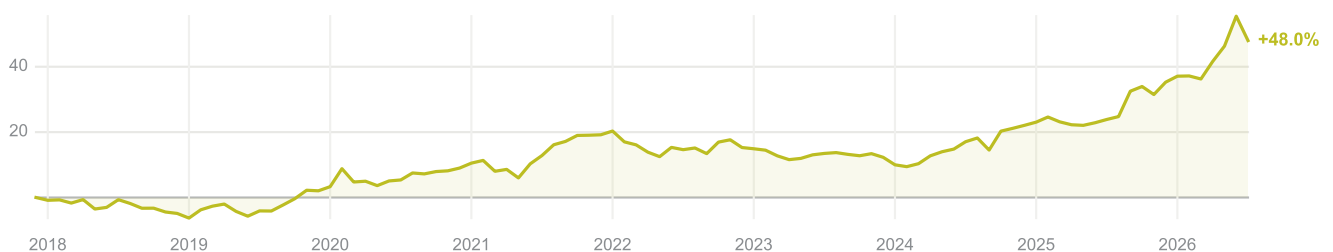
Earnings Quality

CAGR +3.0%

054 ● **Adjusted Interest Coverage Ratio**

Capital Efficiency

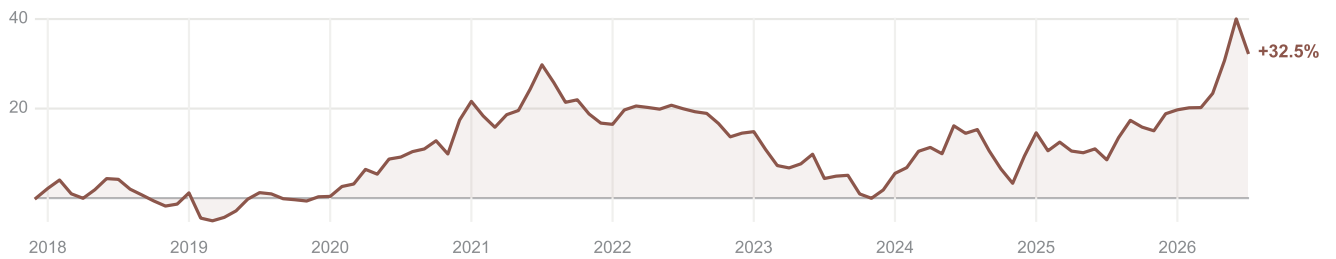
CAGR +2.9%



055 ● 1Y Chg in Cash Flow per Share

Historical Growth

CAGR +2.9%



056 ● 1Y Chg in Free Cash Flow per Share

Historical Growth

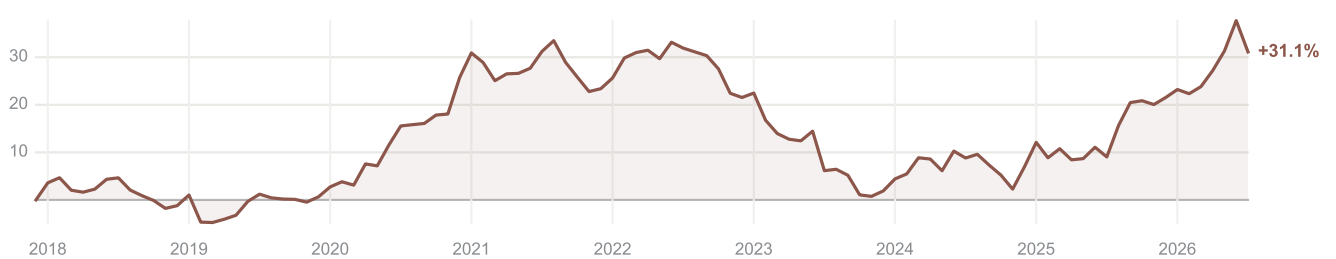
CAGR +2.9%



057 ● 1Y Chg in Cash Flow to Price

Historical Growth

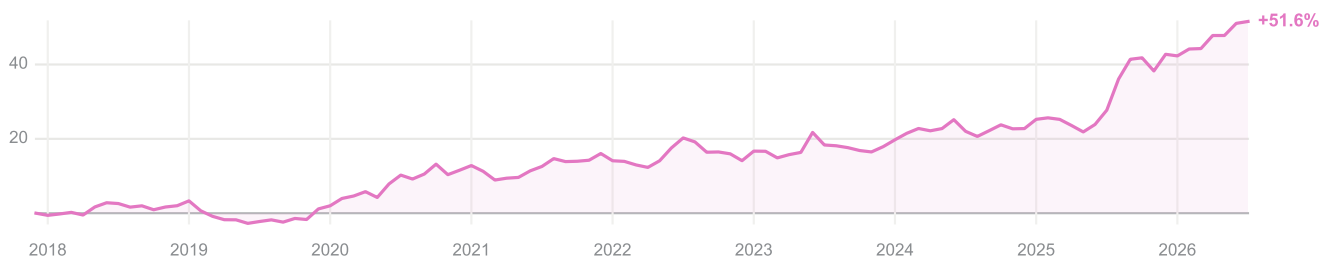
CAGR +2.9%



058 ● Sales to Invested Capital

Earnings Quality

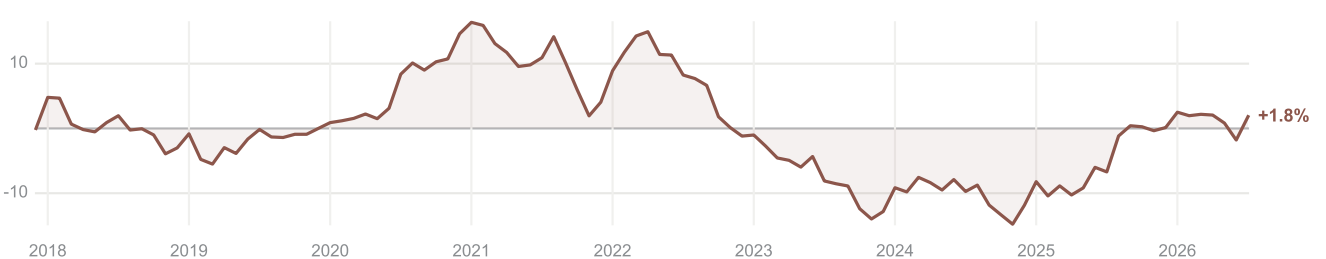
CAGR +2.9%



059 ● 1Y Chg in Earnings to Price

Historical Growth

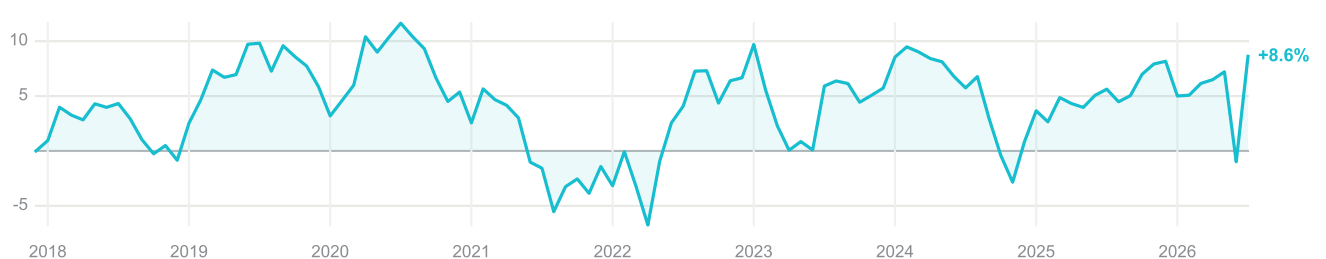
CAGR +2.9%



060 ● 6M Chg in Target Price Gap

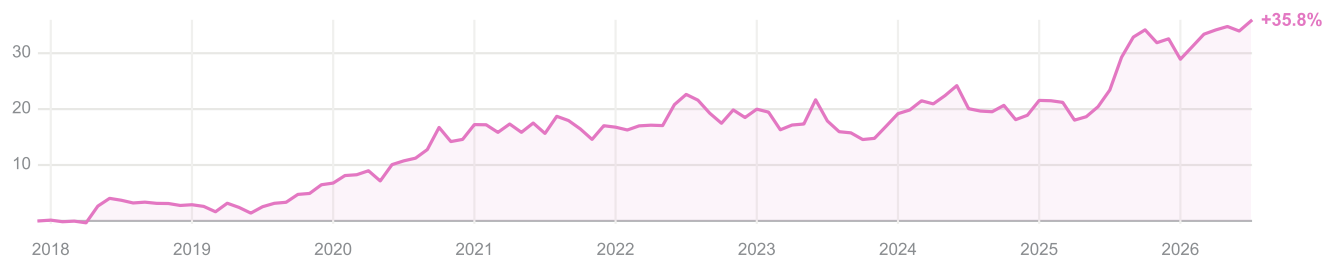
Analyst Expectations

CAGR +2.9%



061 ● Equity Turnover

Earnings Quality CAGR +2.8%



062 ● Net Cash Flow to Price

Valuation CAGR +2.8%



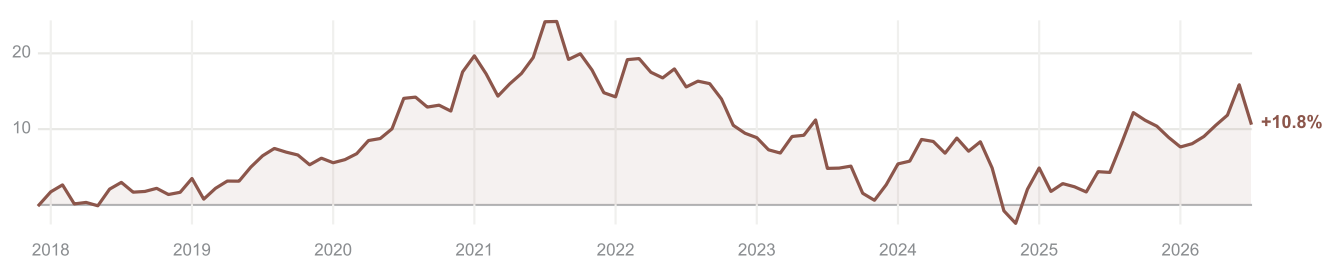
063 ● Cash to Price

Valuation CAGR +2.7%



064 ● 1Y Chg in Operating Margin

Historical Growth CAGR +2.7%



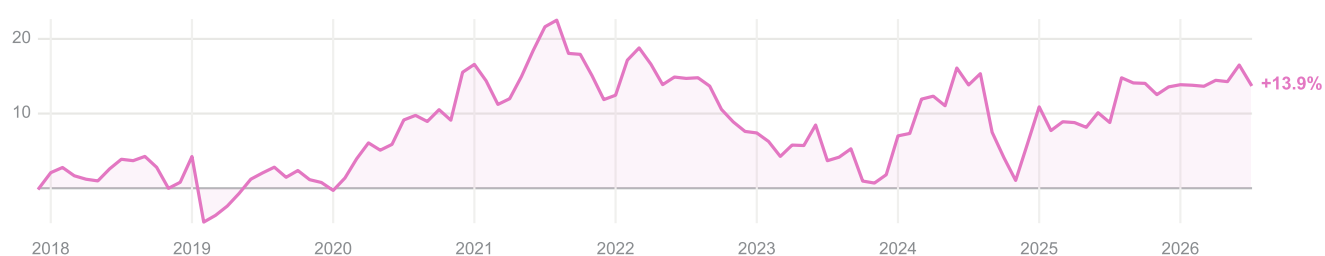
065 ● Cash Flow to Equity

Valuation CAGR +2.7%



066 ● Change EPS to Sales

Earnings Quality CAGR +2.7%



067 ● **Growth Flow to Price**

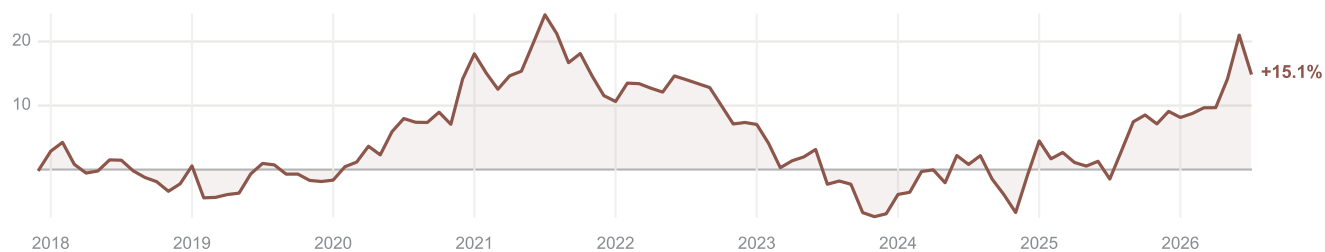
Valuation

CAGR +2.7%

068 ● **1Y Chg in Asset Adjusted Cash Flow**

Historical Growth

CAGR +2.7%

069 ● **1Y Chg in Earnings per Share**

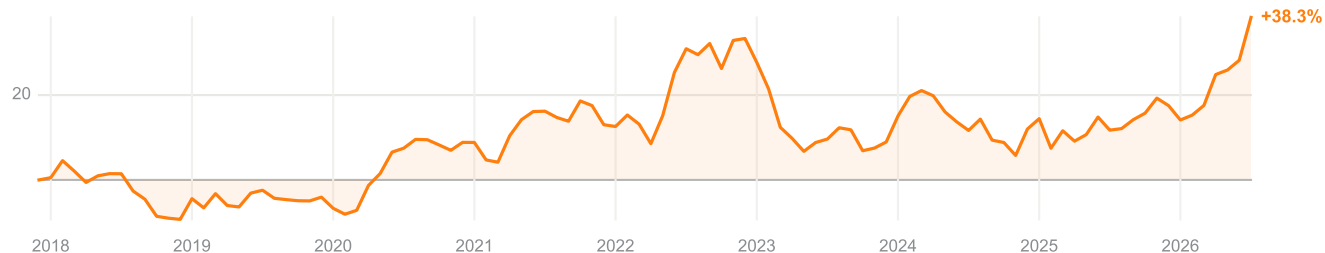
Historical Growth

CAGR +2.6%

070 ● **Risk Adjusted Relative Strength**

Price Momentum

CAGR +2.6%

071 ● **Free Cash Flow to Price**

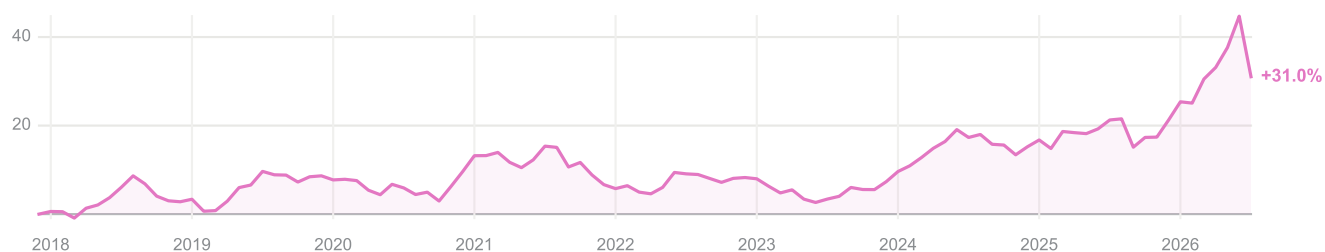
Valuation

CAGR +2.6%

072 ● **Unexpected Receivables Change**

Earnings Quality

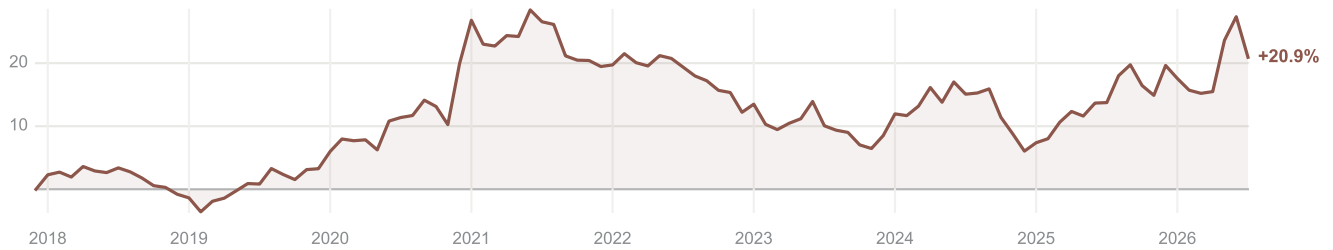
CAGR +2.6%



073 ● 1Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

CAGR +2.6%



074 ● 3Y Chg in Asset Adjusted Cash Flow

Historical Growth

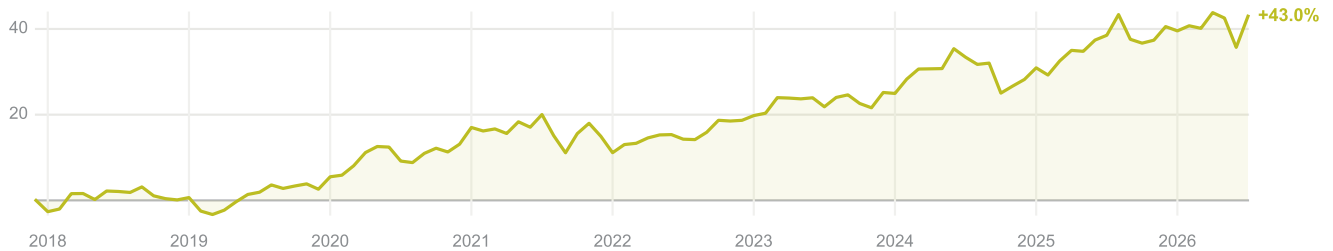
CAGR +2.5%



075 ● Long Term Debt to Assets Ratio

Capital Efficiency

CAGR +2.5%



076 ● 3Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

CAGR +2.5%



077 ● Adjusted EBIT to Price

Valuation

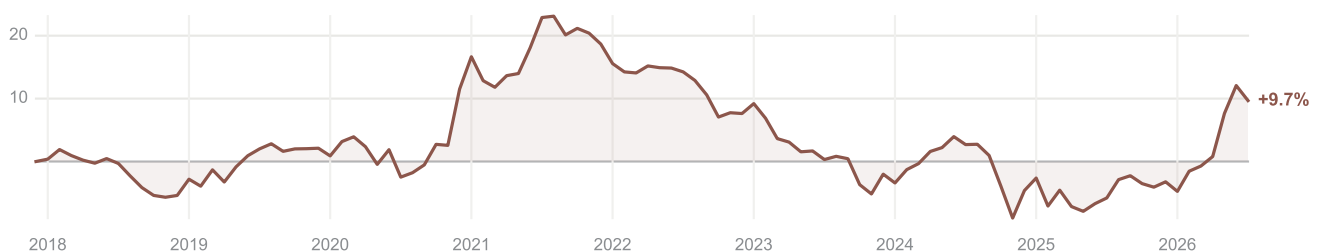
CAGR +2.4%



078 ● 3Y Average Annual Earnings Growth

Historical Growth

CAGR +2.4%



079 ● Sales to EV Ratio

Valuation

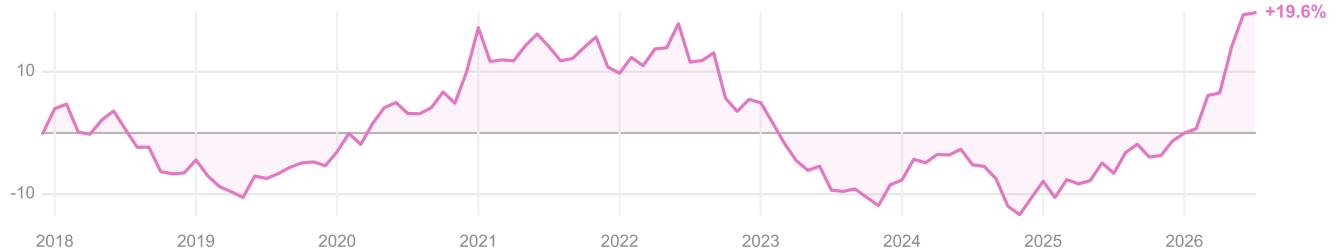
CAGR +2.4%



080 ● Operating Margin

Earnings Quality

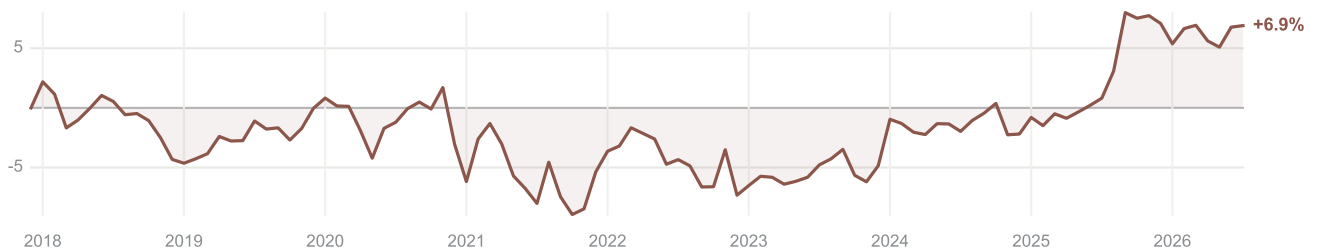
CAGR +2.4%



081 ● 3Y Chg in Operating Cash Flow to Price

Historical Growth

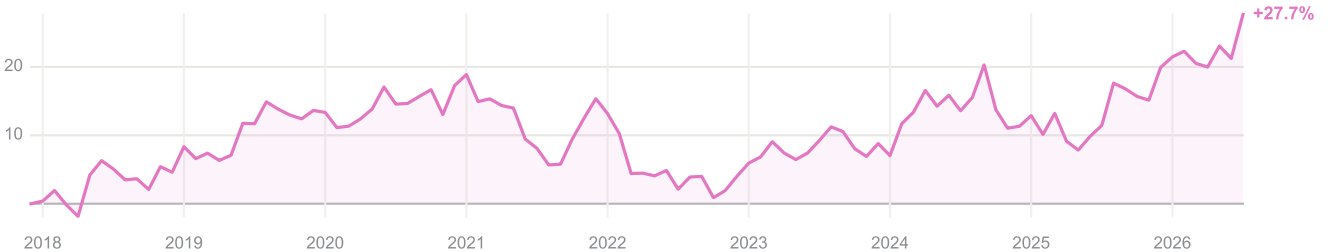
CAGR +2.4%



082 ● Retained Earnings to Total Assets

Earnings Quality

CAGR +2.4%



083 ● Inverse P/E Ratio Adj for Growth and Yield

Valuation

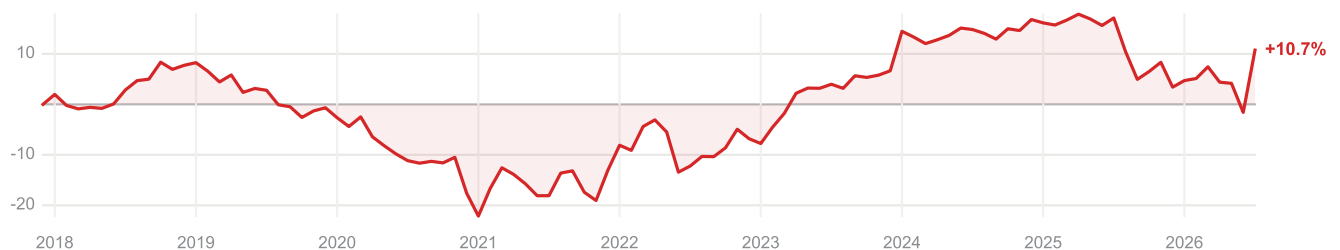
CAGR +2.4%



084 ● Cash to EV Ratio

Valuation

CAGR +2.4%



085 ● **Net Cash Flow to Enterprise Value**

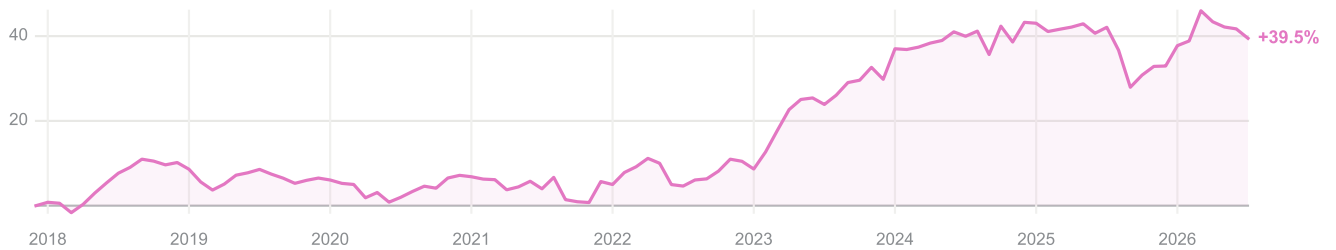
Valuation

CAGR +2.4%

086 ● **Accrual Ratio - Balance Sheet**

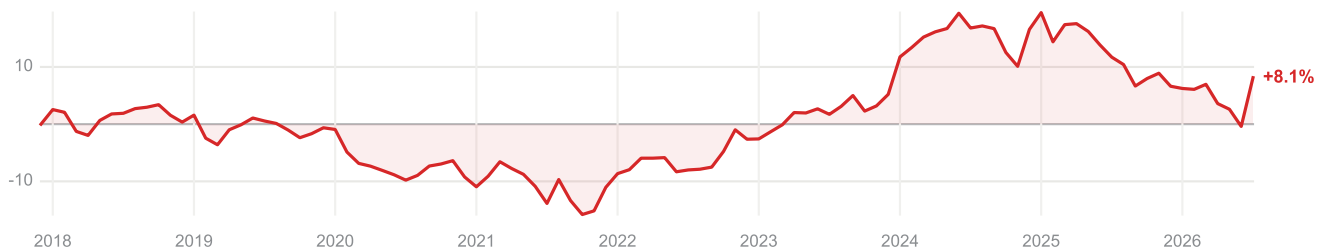
Earnings Quality

CAGR +2.4%

087 ● **Free Cash Flow to Enterprise Value**

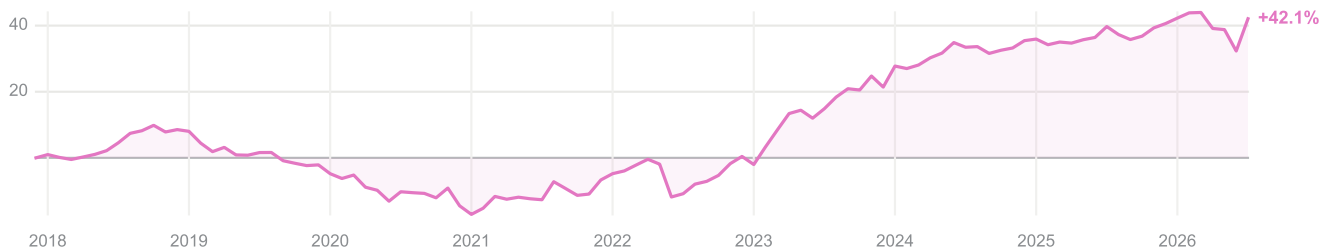
Valuation

CAGR +2.3%

088 ● **Accrual Ratio - Cash Flows**

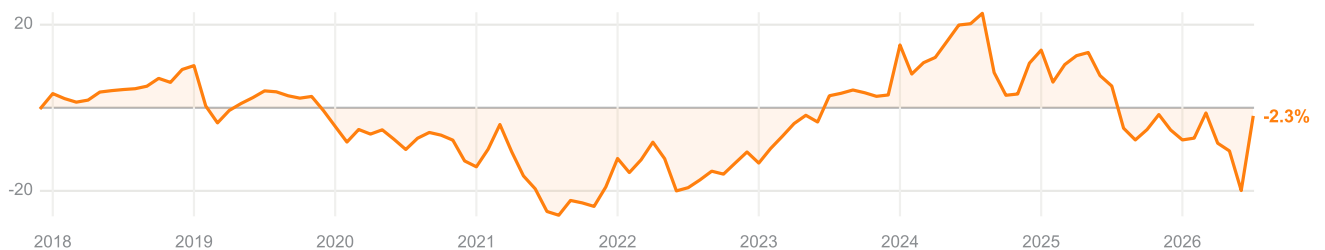
Earnings Quality

CAGR +2.3%

089 ● **Max Daily Return in the Past 6 Months**

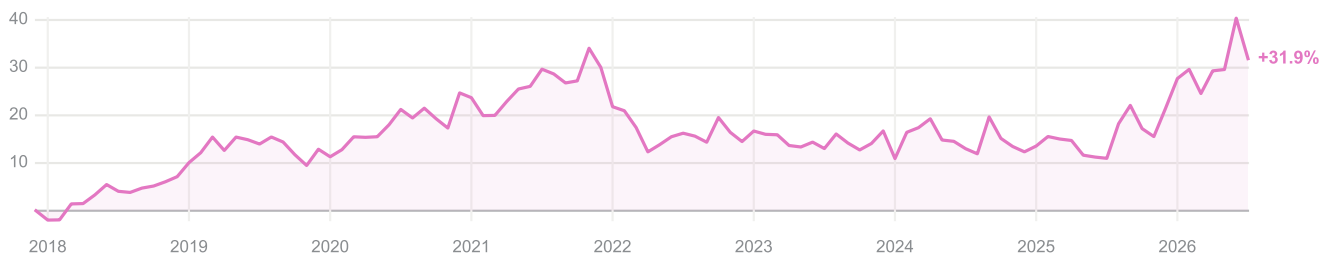
Price Momentum

CAGR +2.3%

090 ● **Working Capital to Assets Ratio**

Earnings Quality

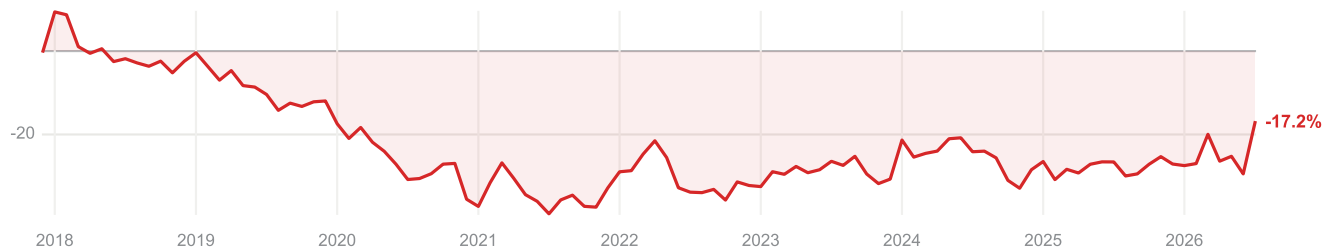
CAGR +2.3%



091 ● **Adjusted Forward Earnings to Price**

Valuation

CAGR +2.3%

092 ● **Log of Unadjusted Stock Price**

Price Momentum

CAGR +2.2%

093 ● **Book to Price**

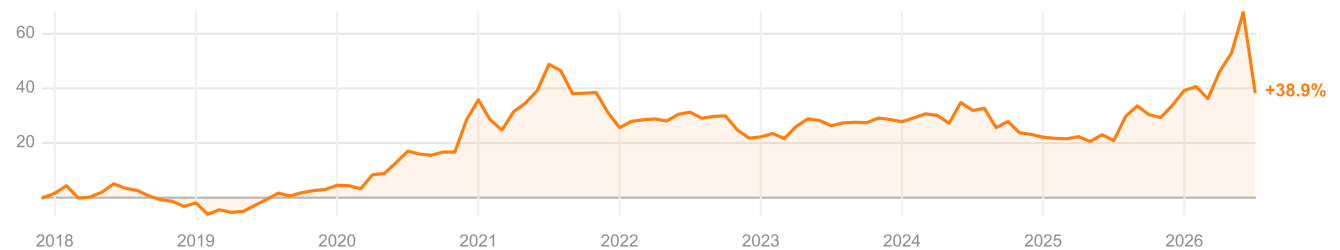
Valuation

CAGR +2.2%

094 ● **Percent Above 260-Day Low**

Price Momentum

CAGR +2.2%

095 ● **Pretax Income to Net Operating Assets**

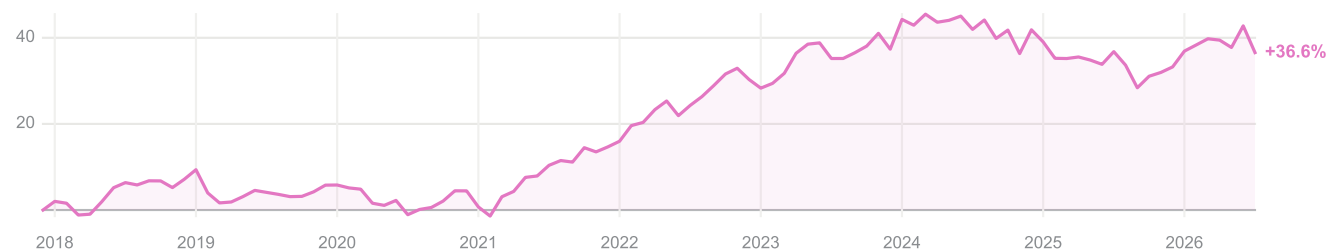
Earnings Quality

CAGR +2.2%

096 ● **Depreciation to Capex Ratio**

Earnings Quality

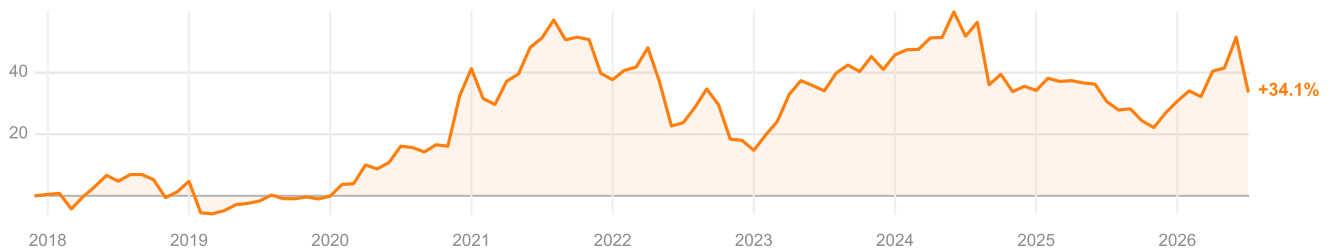
CAGR +2.1%



097 ● **15/36 Week Price Ratio**

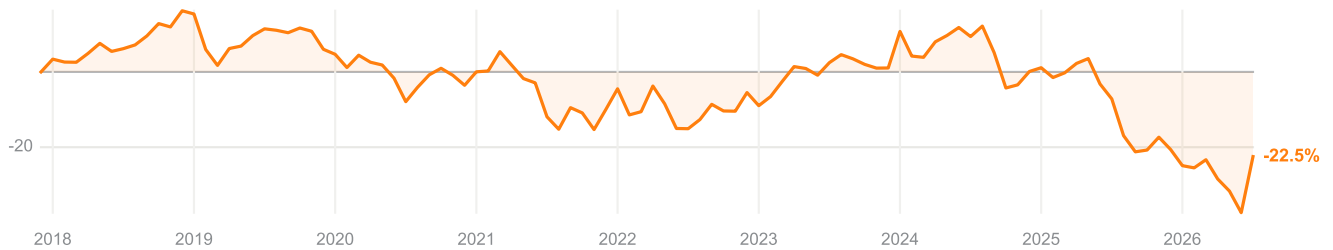
Price Momentum

CAGR +2.1%

098 ● **Adjusted 6 Month Average Share Turn Over**

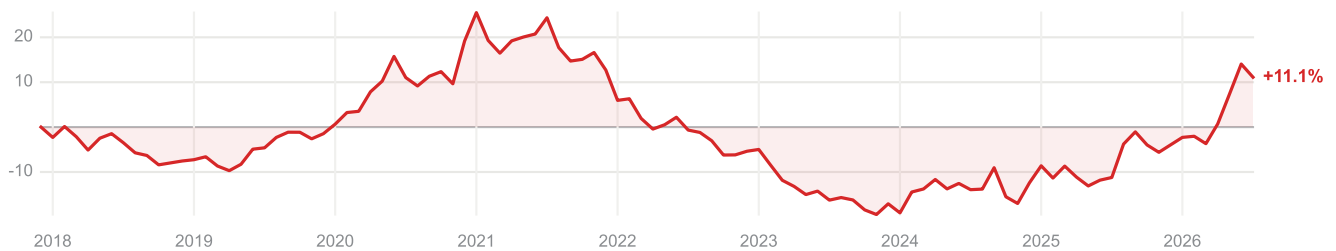
Price Momentum

CAGR +2.1%

099 ● **Solvency Ratio**

Valuation

CAGR +2.1%

100 ● **Cash to Total Assets**

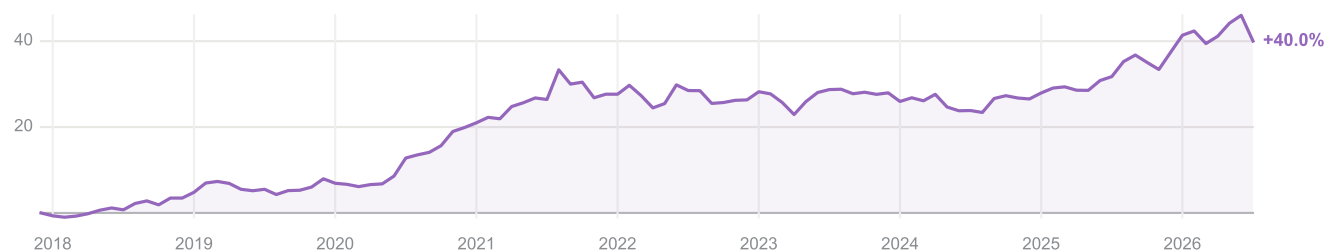
Valuation

CAGR +2.1%

101 ● **60M CAPM Beta**

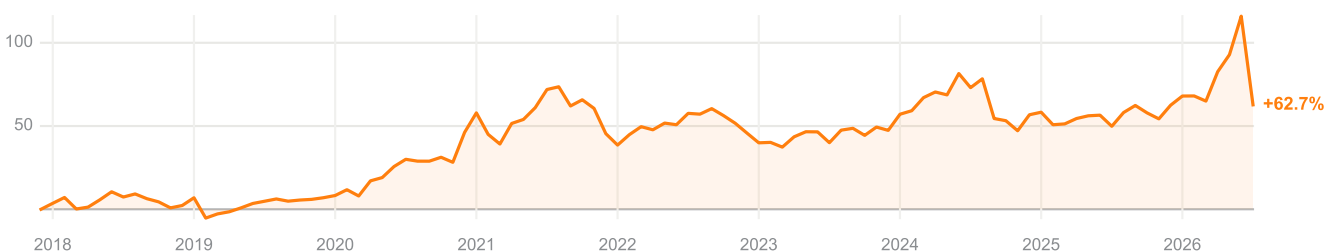
Volatility

CAGR +2.1%

102 ● **12M - 1M Price Momentum**

Price Momentum

CAGR +2.1%



103 ● Operating Earnings to Price Ratio

Valuation

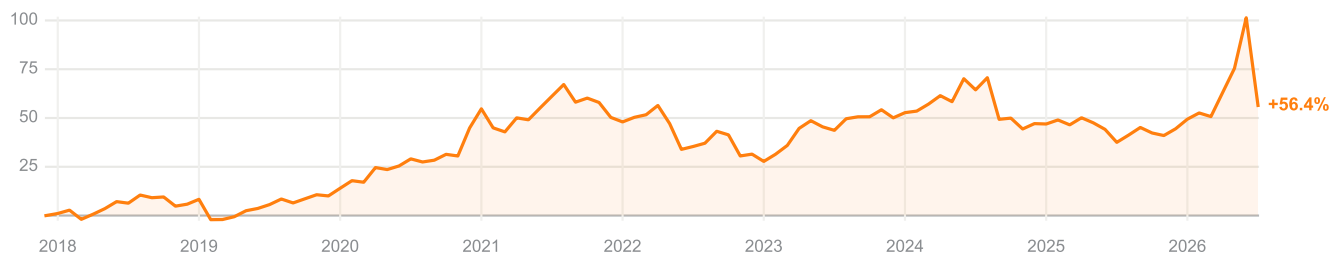
CAGR +2.1%



104 ● 50 Day to 200 Day Stock Price

Price Momentum

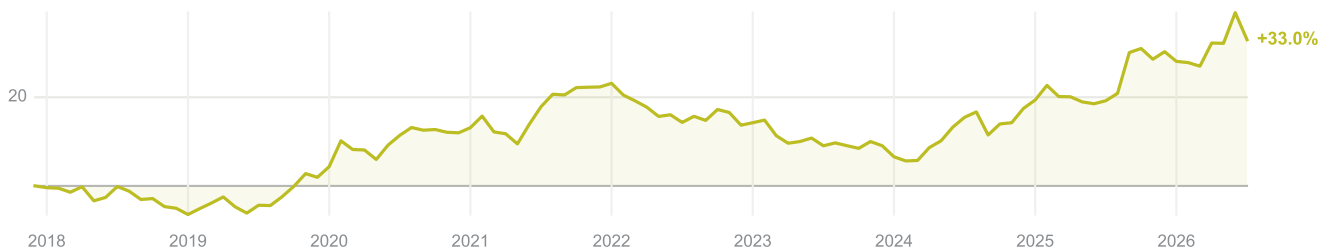
CAGR +2.1%



105 ● Interest Coverage Ratio

Capital Efficiency

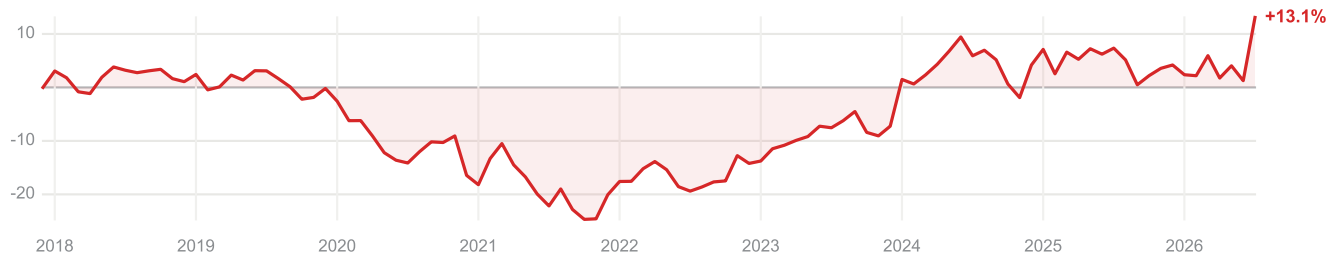
CAGR +2.1%



106 ● Forward Free Cash Flow to Price Ratio

Valuation

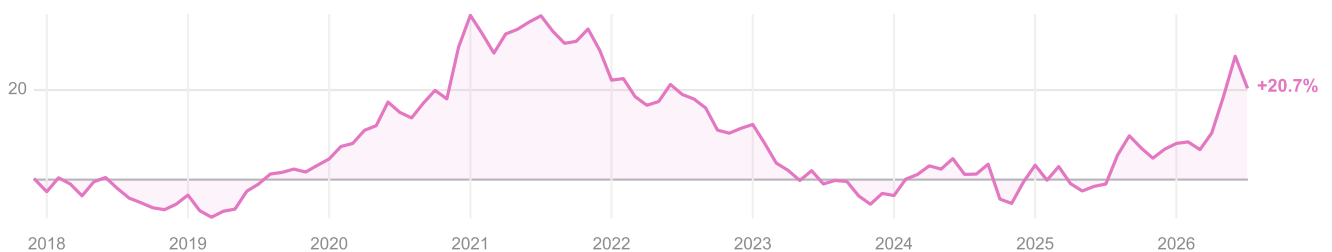
CAGR +2.0%



107 ● Cash Flow to Total Asset

Earnings Quality

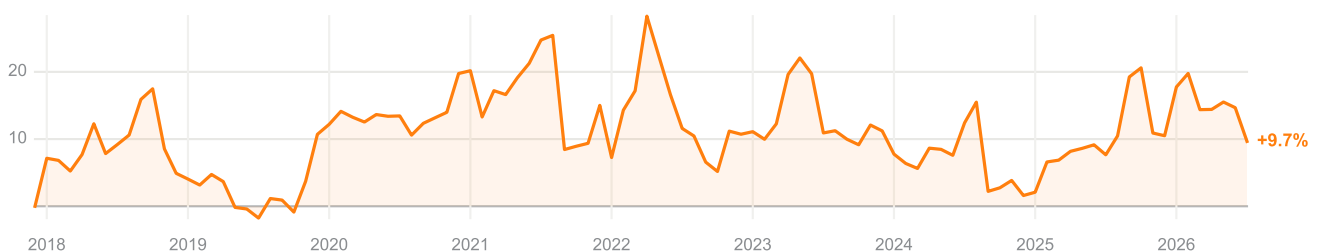
CAGR +2.0%



108 ● 50 Day Volume Signal

Price Momentum

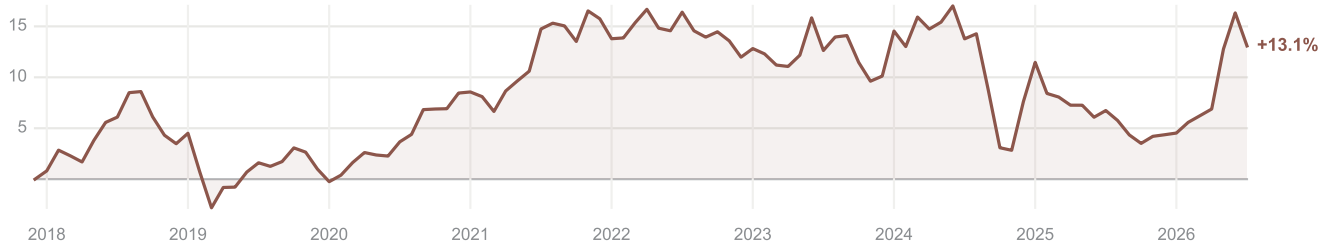
CAGR +2.0%



109 ● 3Y Chg in Sales Turnover

Historical Growth

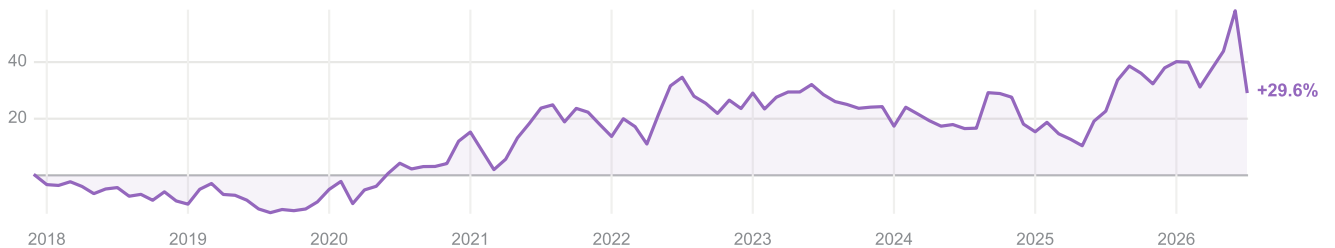
CAGR +2.0%



110 ● 90 Day Coefficient of Variation

Volatility

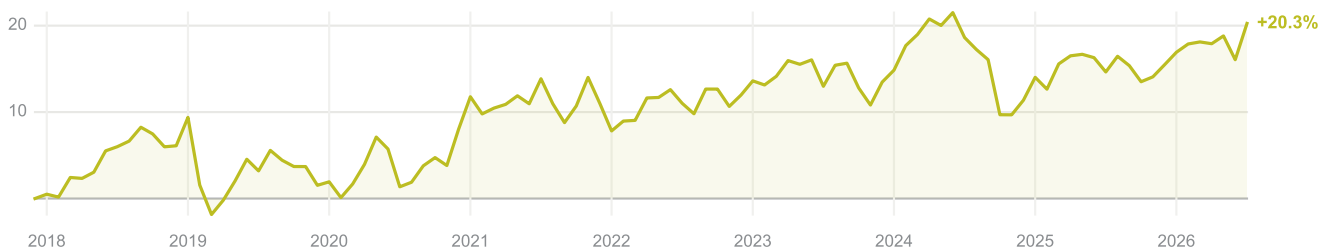
CAGR +2.0%



111 ● Debt to Assets Ratio

Capital Efficiency

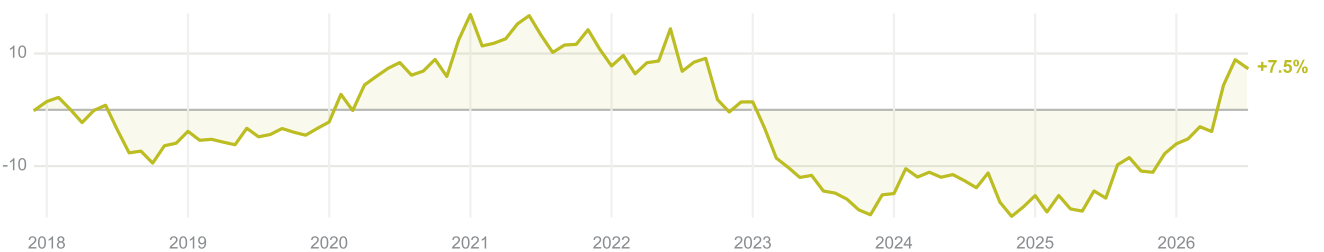
CAGR +2.0%



112 ● Pretax Profit Margin

Capital Efficiency

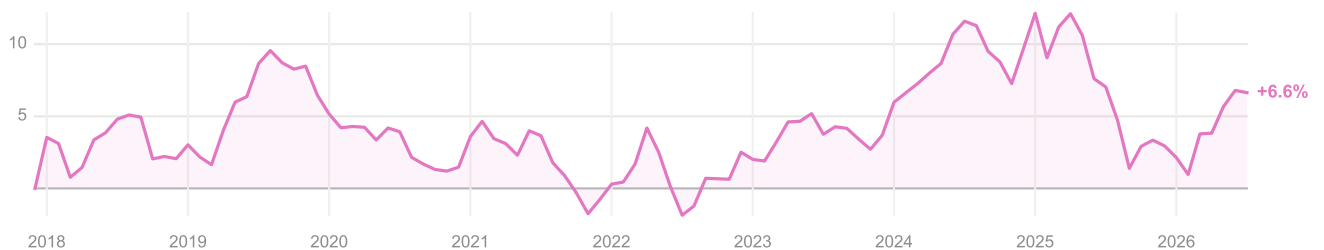
CAGR +2.0%



113 ● Unexpected Inventory Change

Earnings Quality

CAGR +2.0%



114 ● Year over Year Change of Inventory to Assets

Earnings Quality

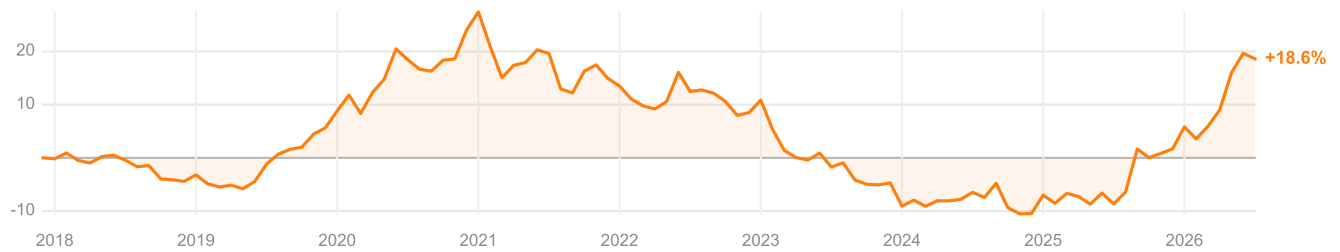
CAGR +1.9%



115 ● Institution Ownership Level

Price Momentum

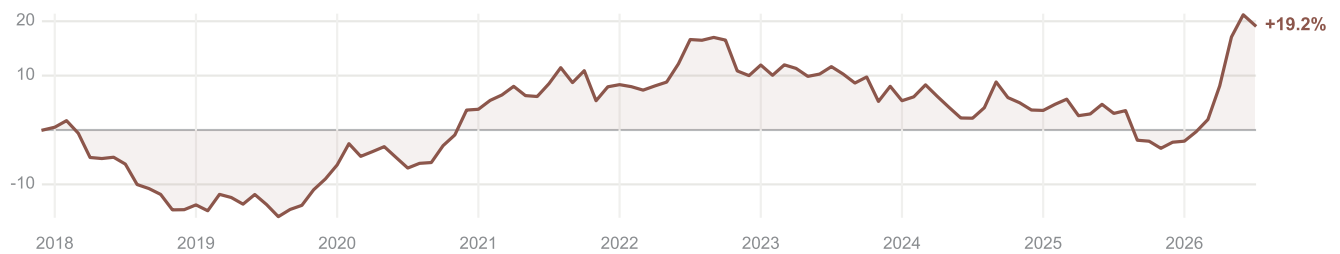
CAGR +1.9%



116 ● Dividend Growth

Historical Growth

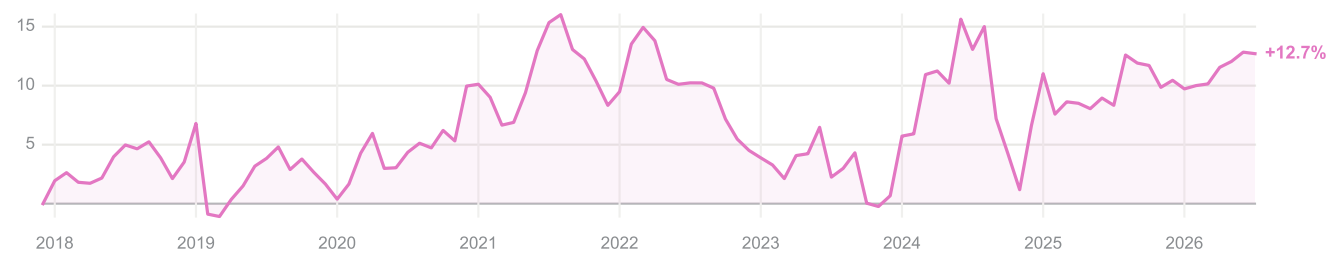
CAGR +1.9%



117 ● Year over Year Change of EPS to Sales

Earnings Quality

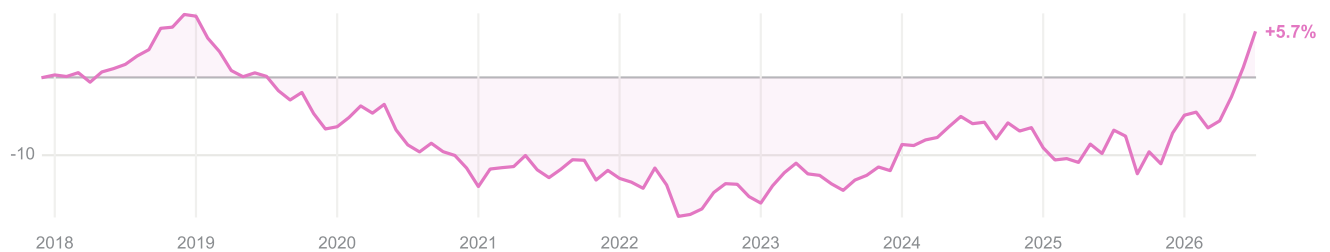
CAGR +1.9%



118 ● Change in TTM Depr. to Capex

Earnings Quality

CAGR +1.9%



119 ● 3Y Chg in Asset Adjusted EPS

Historical Growth

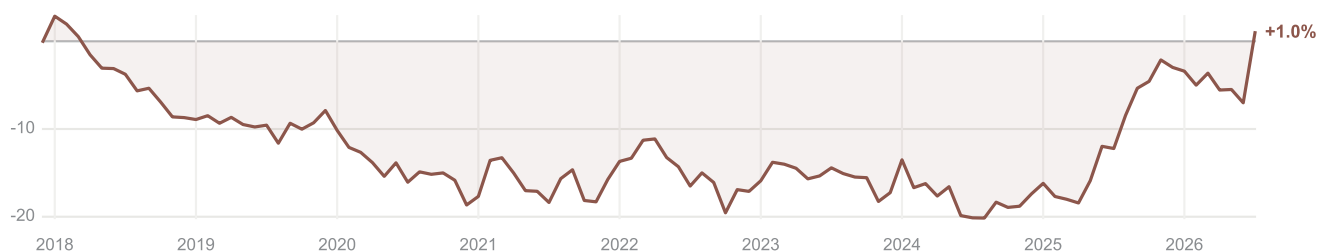
CAGR +1.8%



120 ● 3Y Chg in Cash Flow to Price

Historical Growth

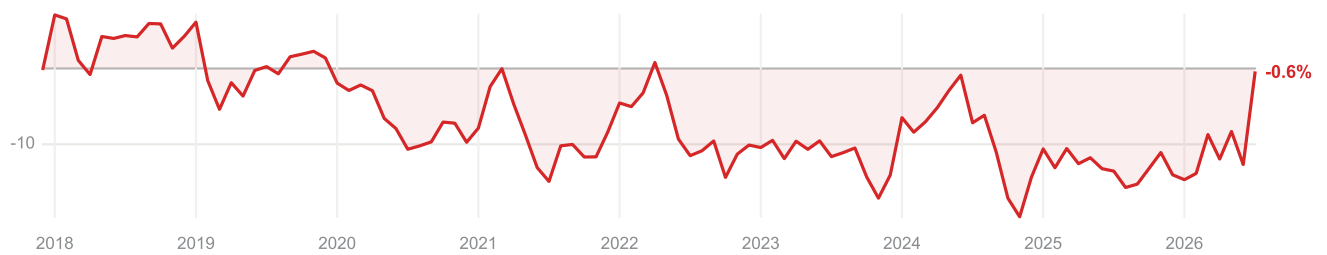
CAGR +1.8%



121 ● Pretax Income Per Share to Price

Valuation

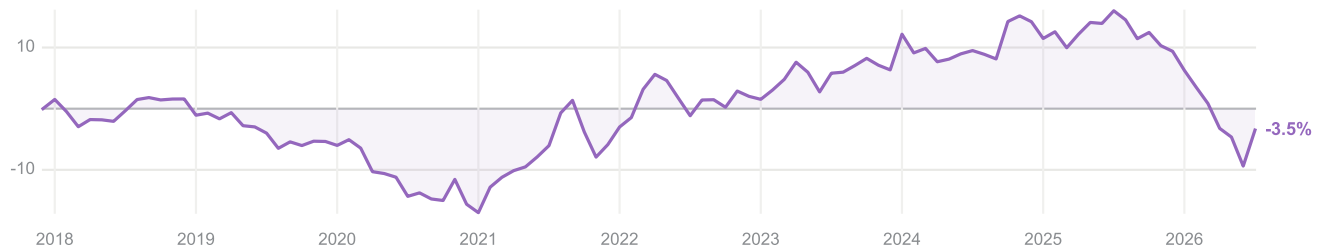
CAGR +1.8%



122 ● Book Leverage

Volatility

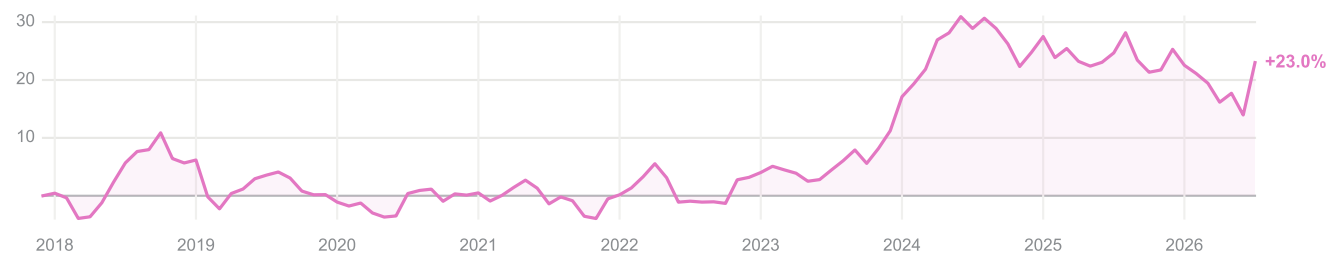
CAGR +1.8%



123 ● Cash Burn Rate

Earnings Quality

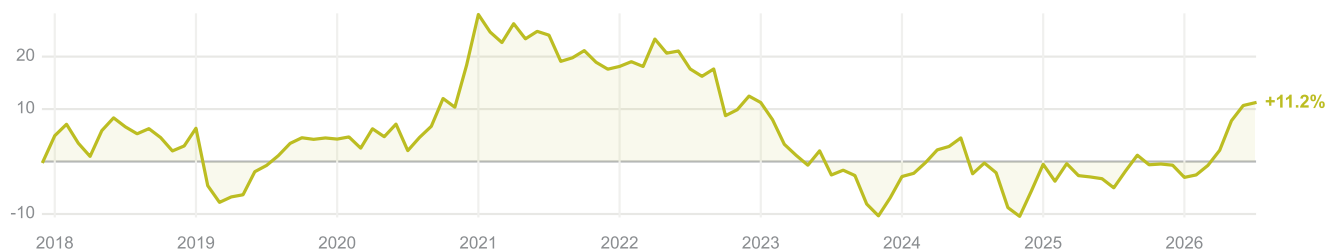
CAGR +1.8%



124 ● Return on Equity

Capital Efficiency

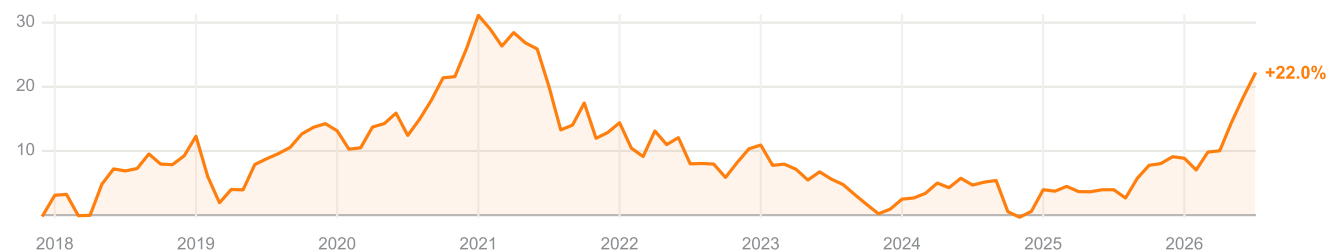
CAGR +1.8%



125 ● Foreign Institution Ownership

Price Momentum

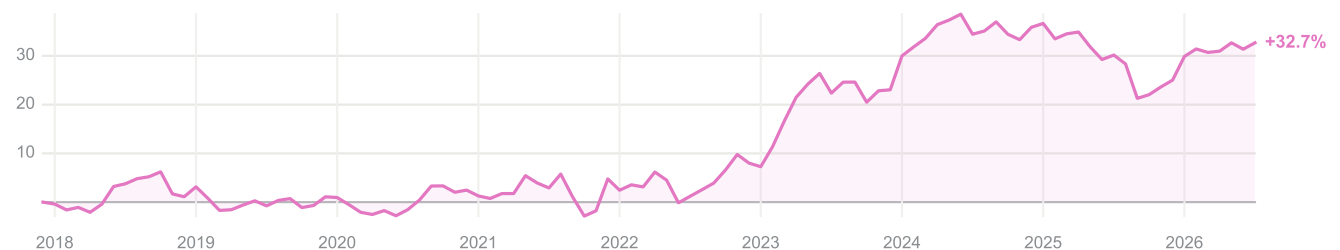
CAGR +1.8%



126 ● Capital Expenditure to Sales

Earnings Quality

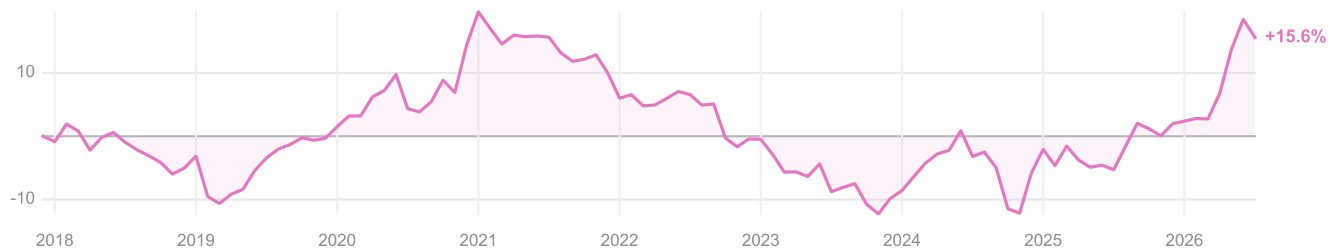
CAGR +1.8%



127 ● **Cash Flow to Invested Capital**

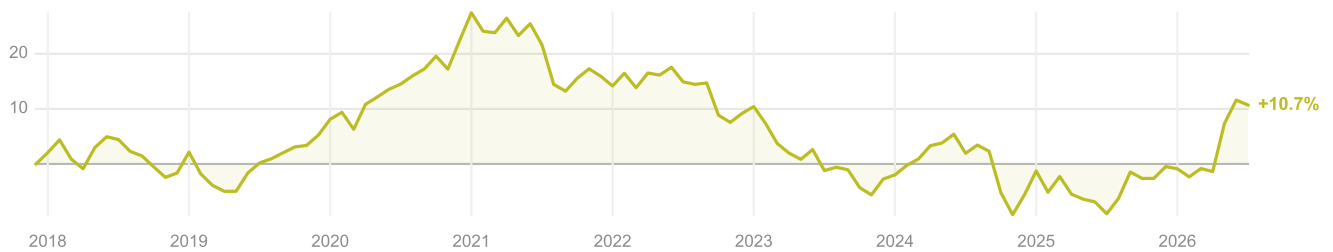
Earnings Quality

CAGR +1.8%

128 ● **EBITDA to Assets**

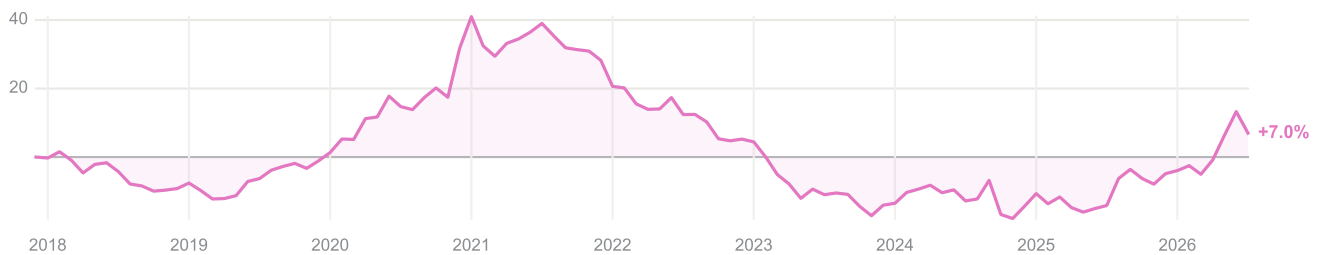
Capital Efficiency

CAGR +1.7%

129 ● **Operating Earnings to Assets Ratio**

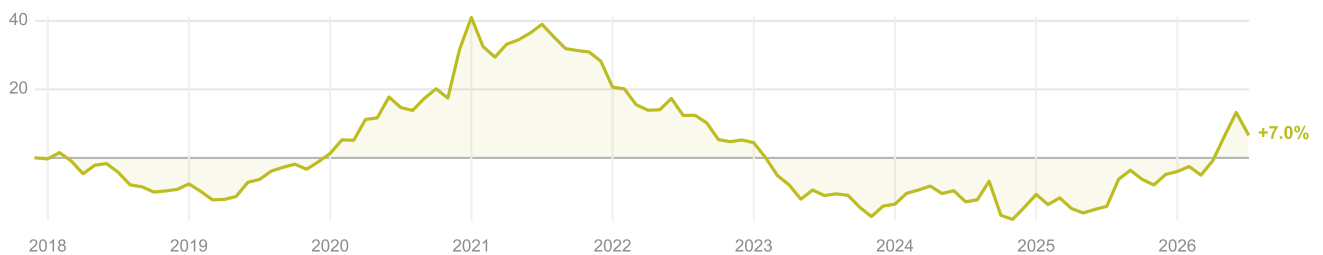
Earnings Quality

CAGR +1.7%

130 ● **Return on Assets**

Capital Efficiency

CAGR +1.7%

131 ● **3Y Average Annual Sales Growth**

Historical Growth

CAGR +1.7%

132 ● **Net Current Assets to Price Ratio**

Valuation

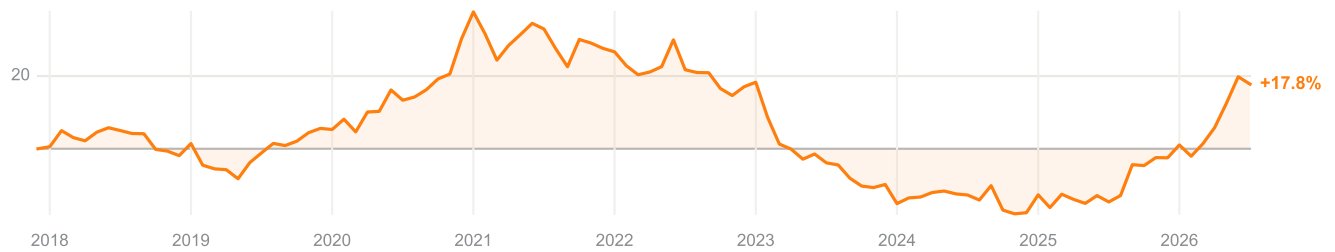
CAGR +1.7%



133 ● Institution Ownership Level - Active Manager

Price Momentum

CAGR +1.7%



134 ● Log of Market Capitalization Cubed

Size

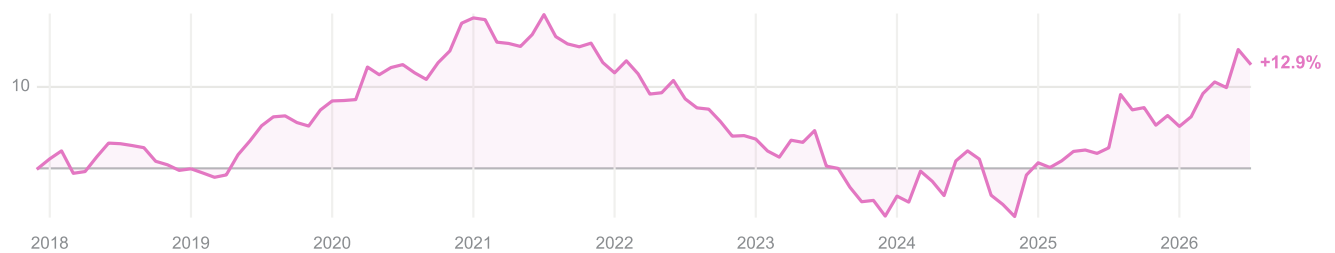
CAGR +1.7%



135 ● Year over Year Change in SGA to Sales

Earnings Quality

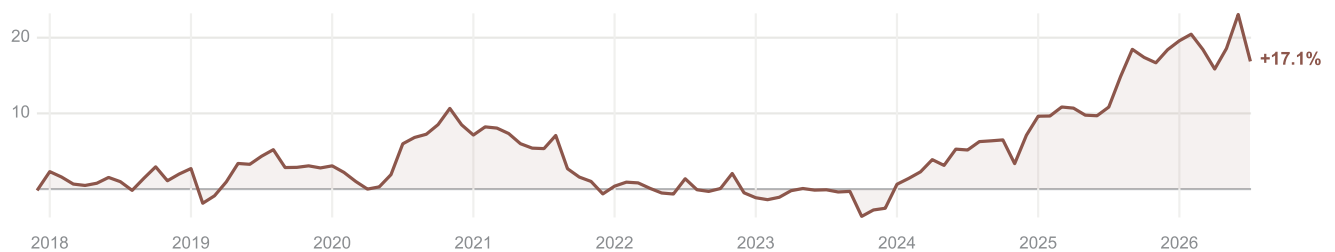
CAGR +1.7%



136 ● 3Y Chg in Free Cash Flow to Price

Historical Growth

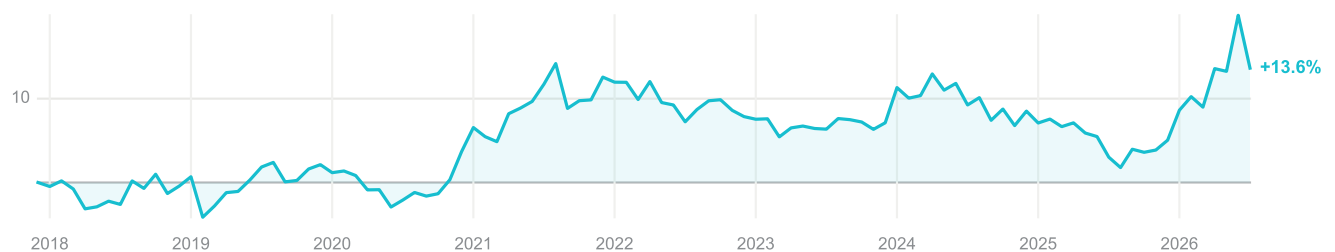
CAGR +1.7%



137 ● Buy-to-Sell Recommendation Ratio Less 3M SMA

Analyst Expectations

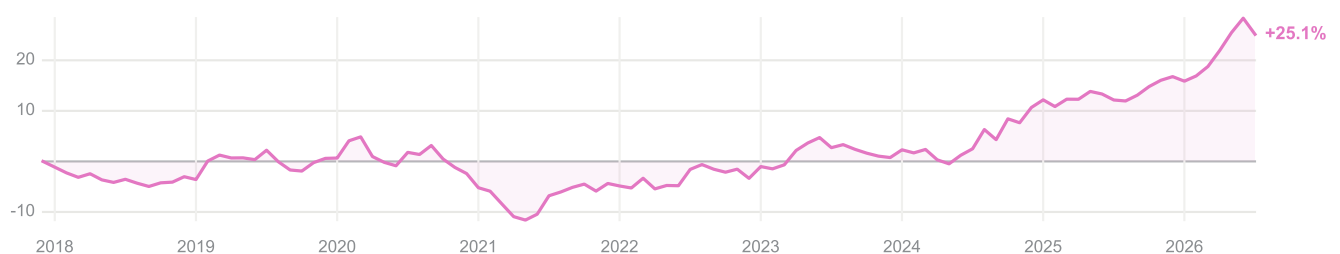
CAGR +1.6%



138 ● SG&A to Sales

Earnings Quality

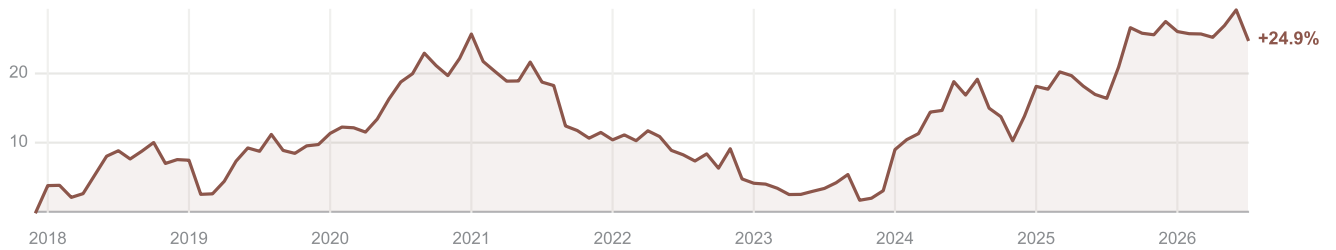
CAGR +1.6%



139 ● **3Y Chg in Asset Adjusted Free Cash Flow**

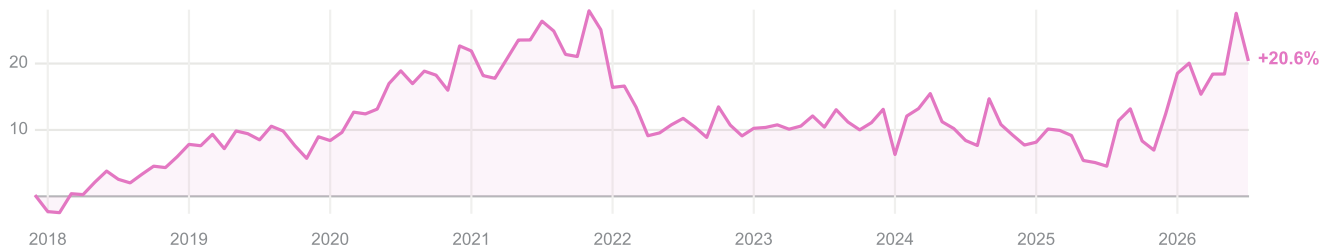
Historical Growth

CAGR +1.6%

140 ● **Working Capital to Total Assets**

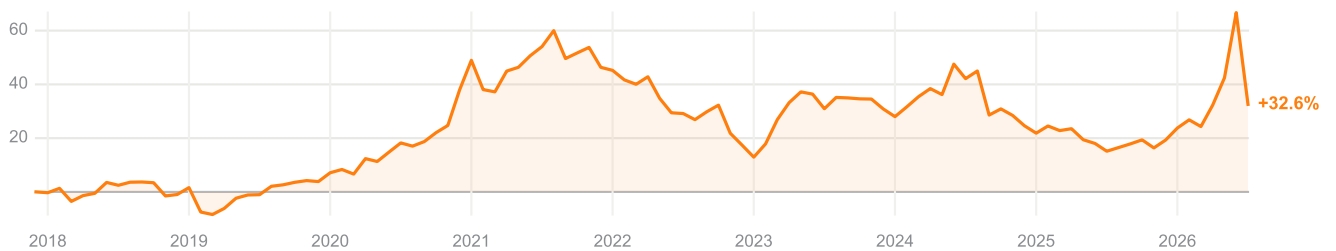
Earnings Quality

CAGR +1.6%

141 ● **26W Price Stat**

Price Momentum

CAGR +1.6%

142 ● **Dividends to Price Ratio**

Valuation

CAGR +1.5%

143 ● **Return on Net Tangible Assets**

Valuation

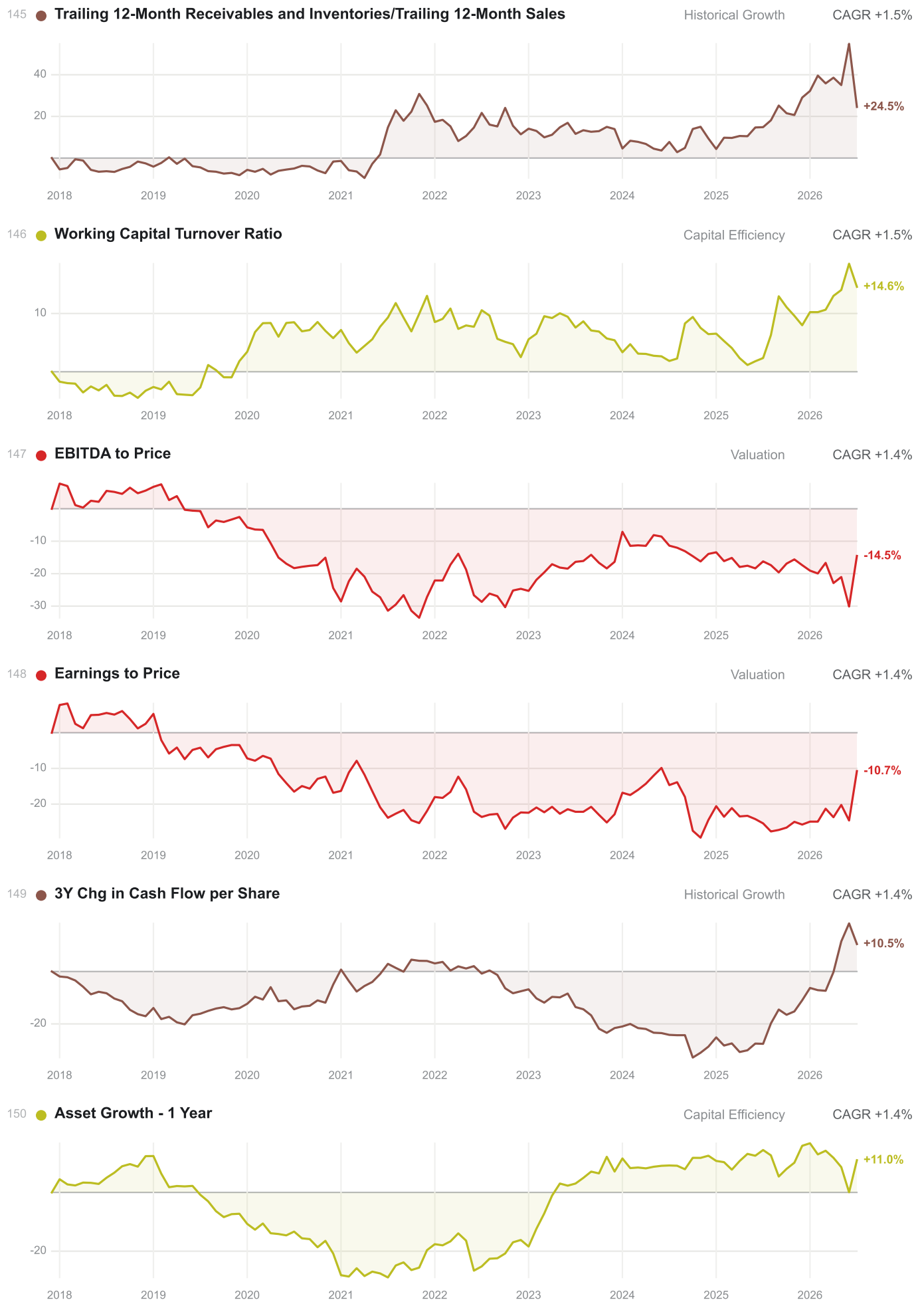
CAGR +1.5%

144 ● **Log Market Cap**

Size

CAGR +1.5%





151 ● **6 Months Price Reversal**

Price Momentum

CAGR +1.3%

152 ● **Ownership Breadth Stability of Institution Ownership by Buyers**

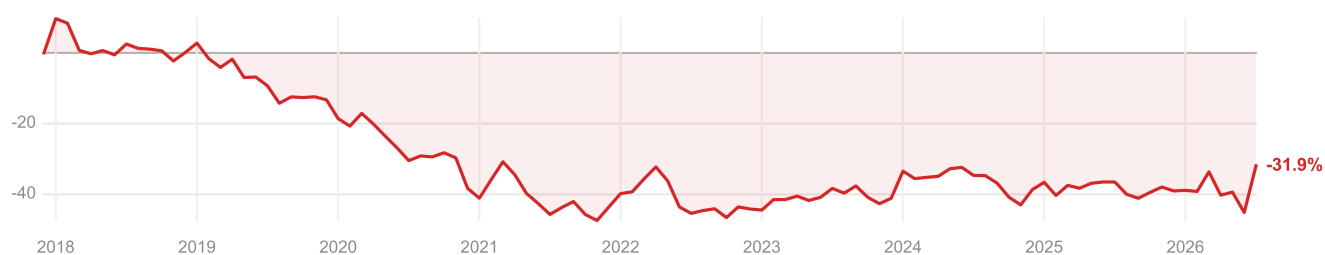
Price Momentum

CAGR +1.3%

153 ● **Forward Earnings to Price**

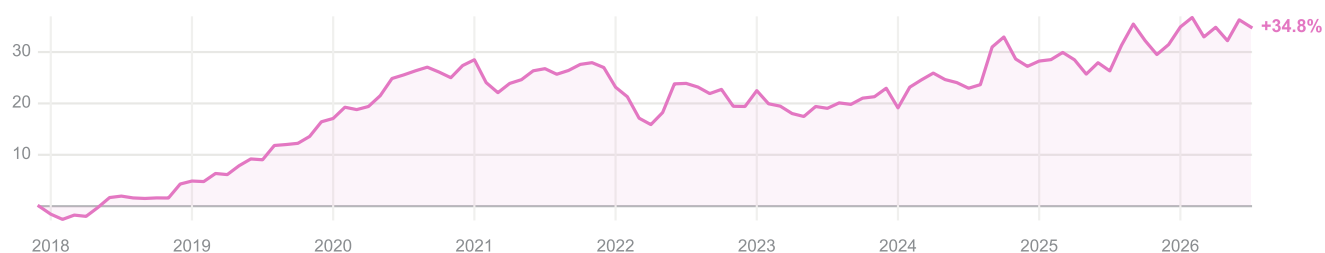
Valuation

CAGR +1.3%

154 ● **Cash Ratio**

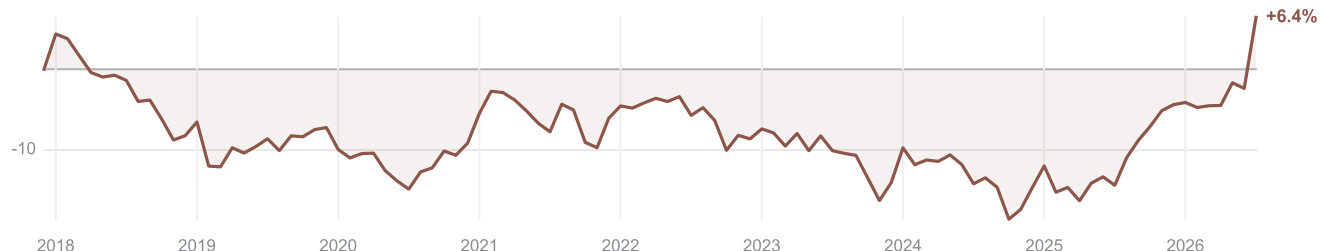
Earnings Quality

CAGR +1.3%

155 ● **3Y Chg in Earnings to Price**

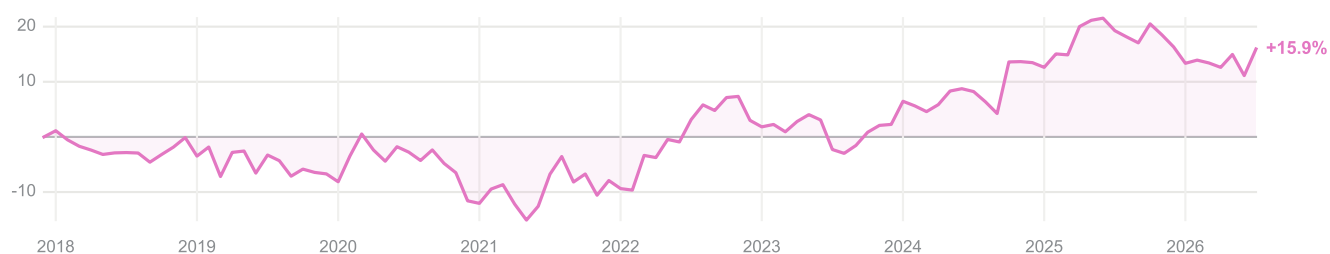
Historical Growth

CAGR +1.3%

156 ● **Working Capital to Sales**

Earnings Quality

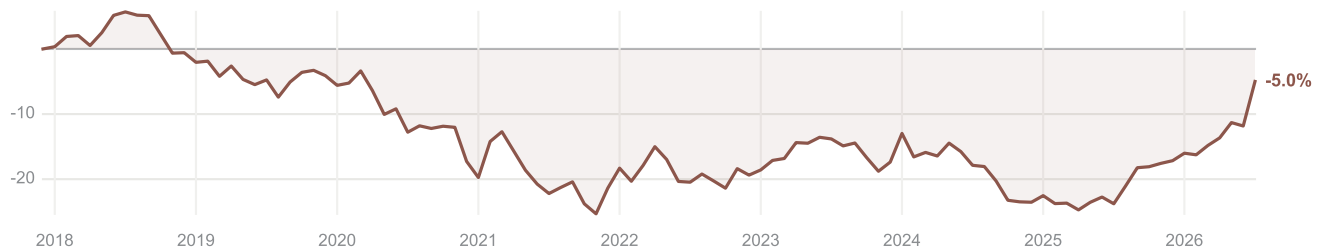
CAGR +1.3%



157 ● 3Y Chg in Sales to Price

Historical Growth

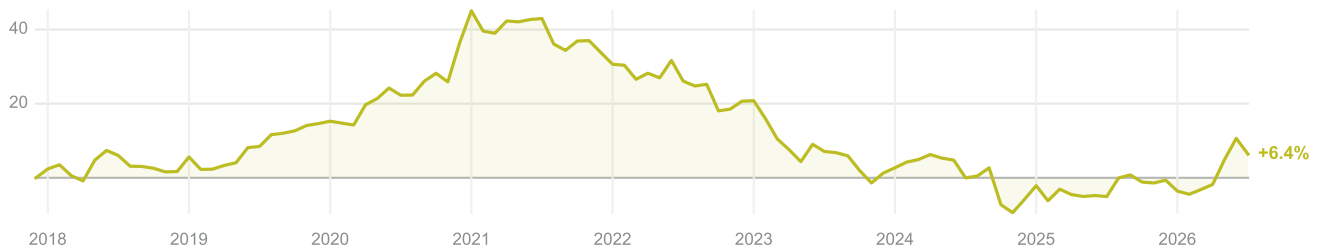
CAGR +1.3%



158 ● Return on Invested Capital

Capital Efficiency

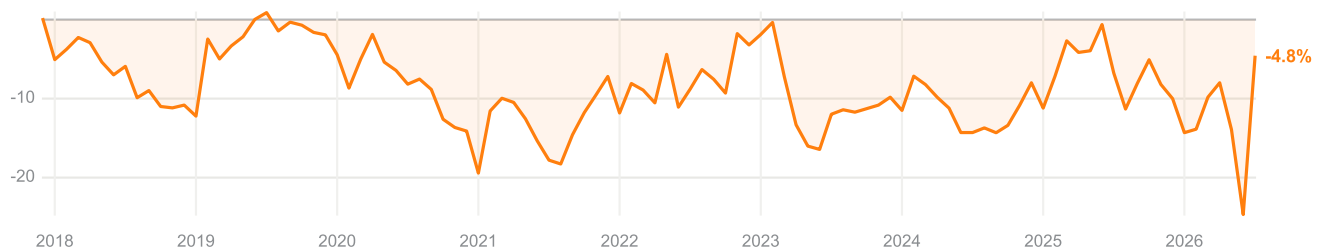
CAGR +1.2%



159 ● 60M CAPM Alpha

Price Momentum

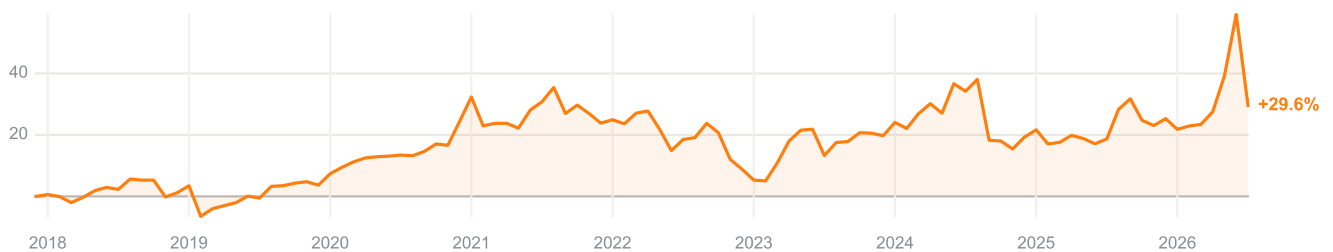
CAGR +1.2%



160 ● Closing Price to 52 Week High

Price Momentum

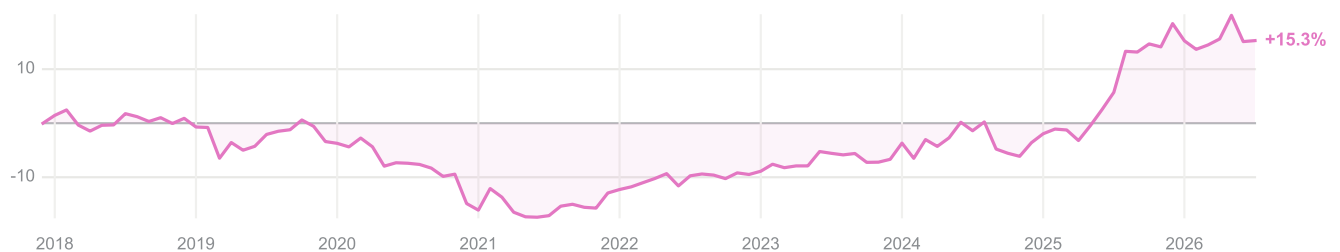
CAGR +1.2%



161 ● Cash to Sales

Earnings Quality

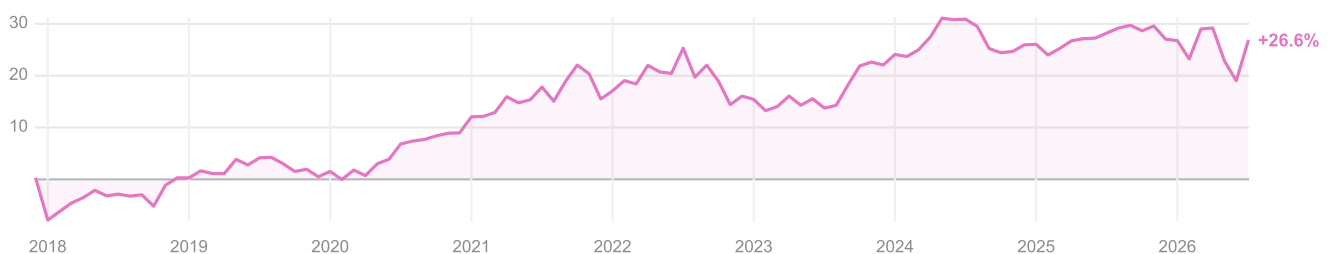
CAGR +1.2%

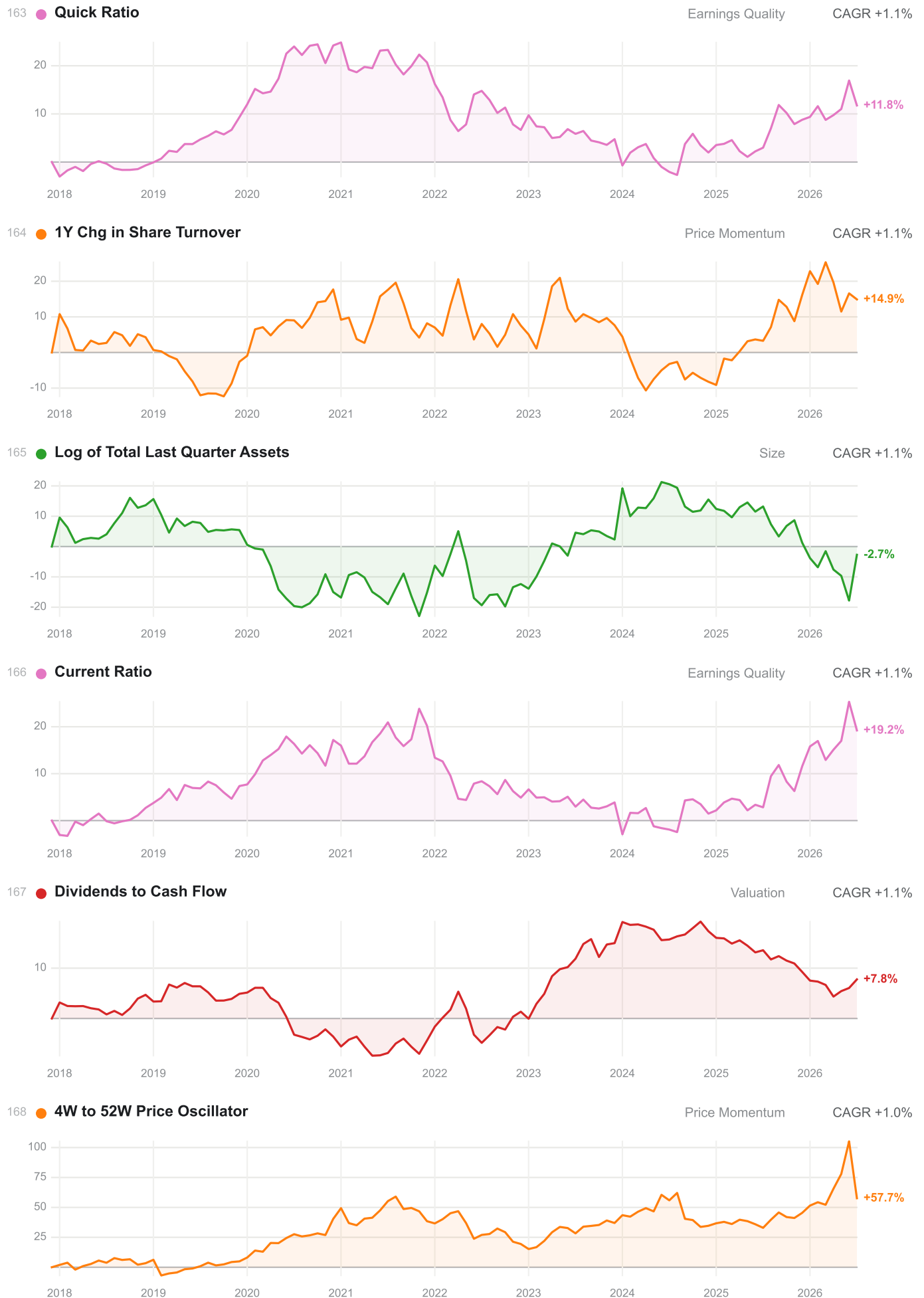


162 ● Inventory to Assets Ratio

Earnings Quality

CAGR +1.2%

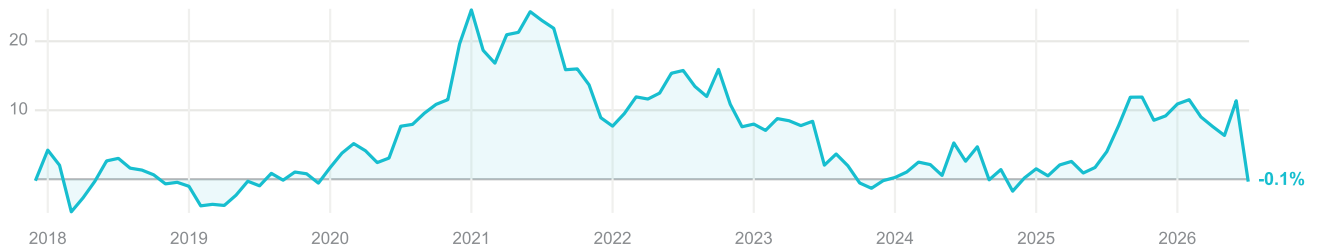




169 ● **Adj Number of EPS FY2 Revisions**

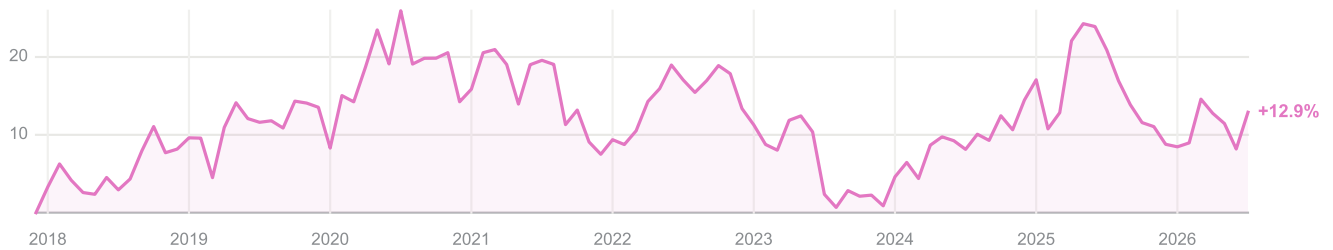
Analyst Expectations

CAGR +1.0%

170 ● **Inventory Turnover**

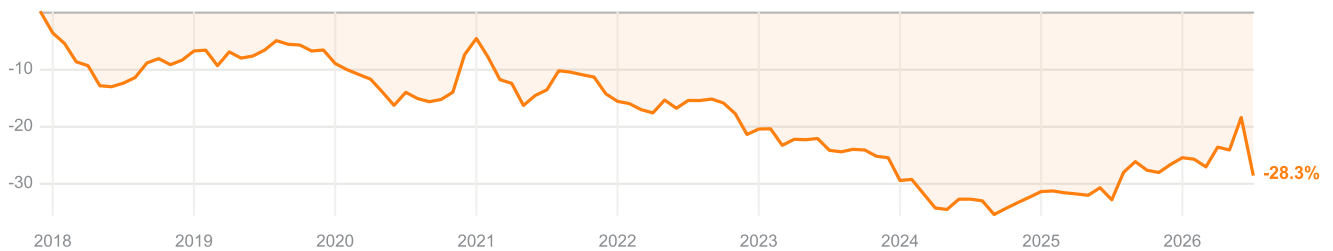
Earnings Quality

CAGR +1.0%

171 ● **6 Month Average Share Turn Over**

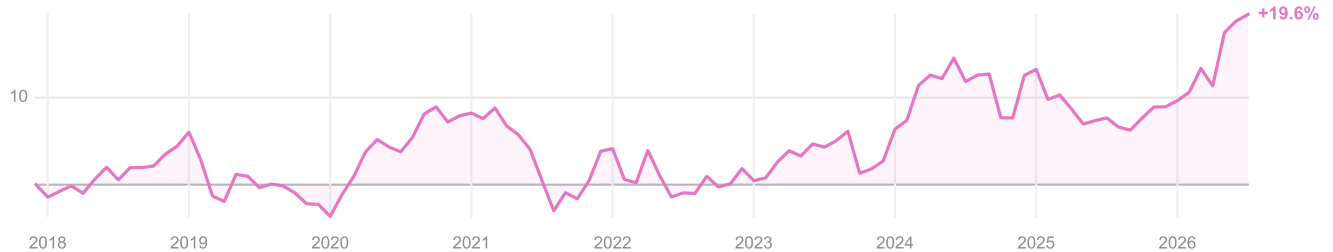
Price Momentum

CAGR +1.0%

172 ● **Change of Sales to Change of EPS**

Earnings Quality

CAGR +1.0%

173 ● **EBITDA to Enterprise Value**

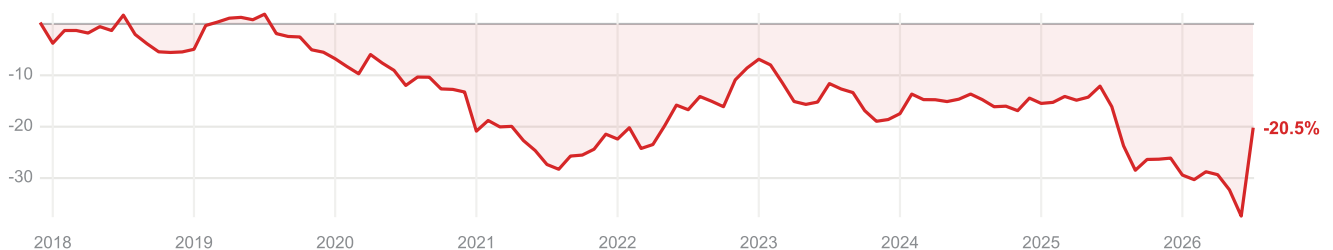
Valuation

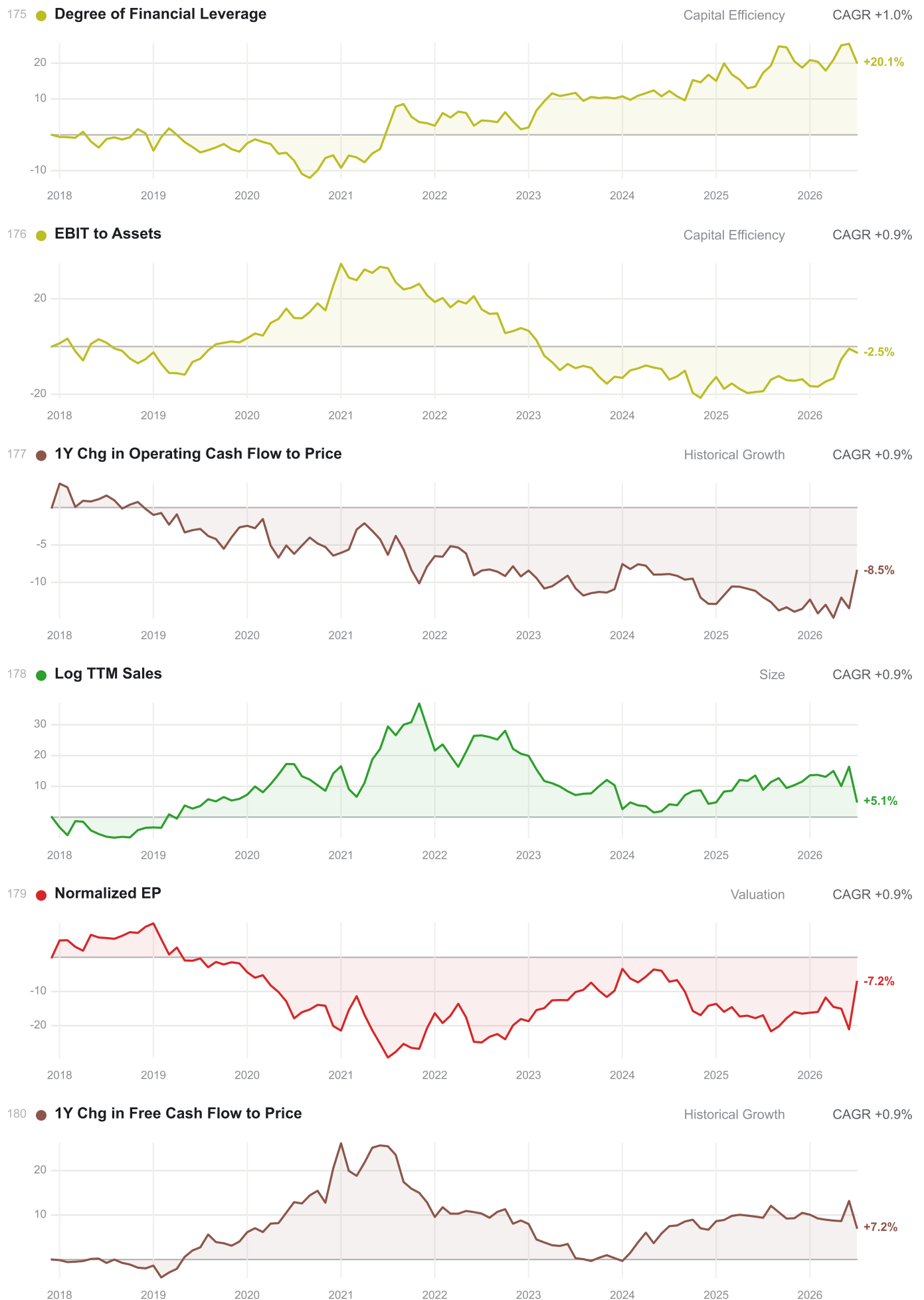
CAGR +1.0%

174 ● **8-Week Change in 12-Month Forward Earnings Consensus Estimate/Price**

Valuation

CAGR +1.0%

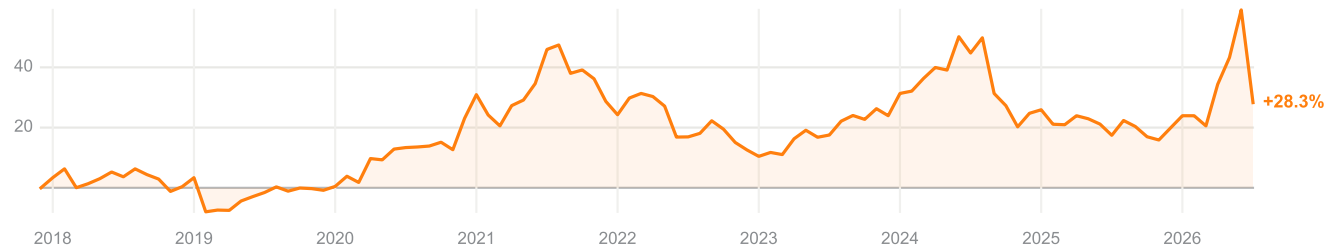




181 ● 39W Lagged Return

Price Momentum

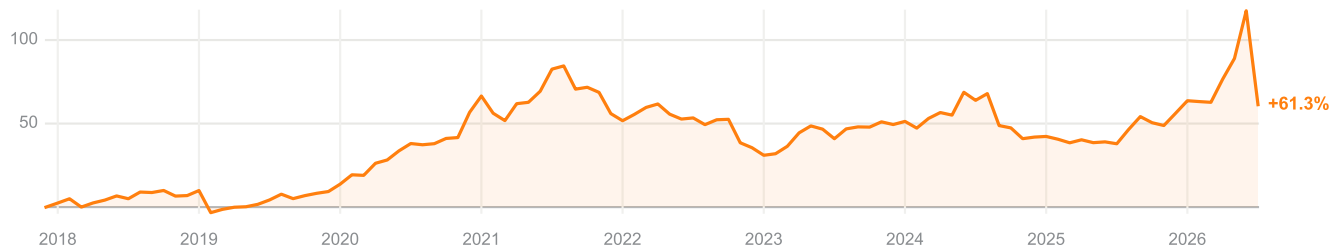
CAGR +0.8%



182 ● 52W High Low

Price Momentum

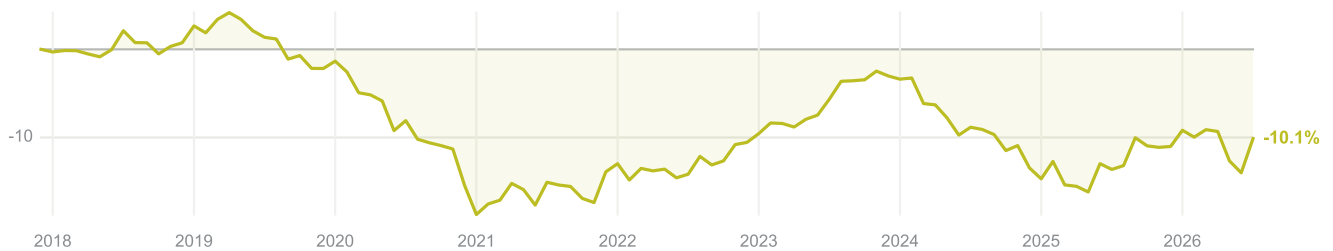
CAGR +0.8%



183 ● 1Y Chg in Shares Outstanding

Capital Efficiency

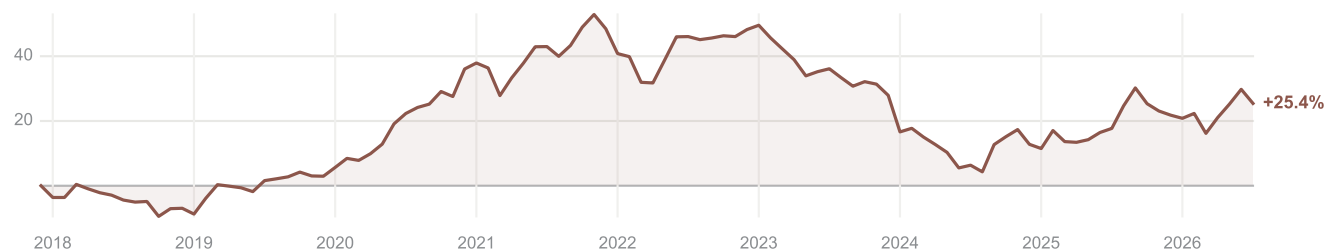
CAGR +0.8%



184 ● Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1

Historical Growth

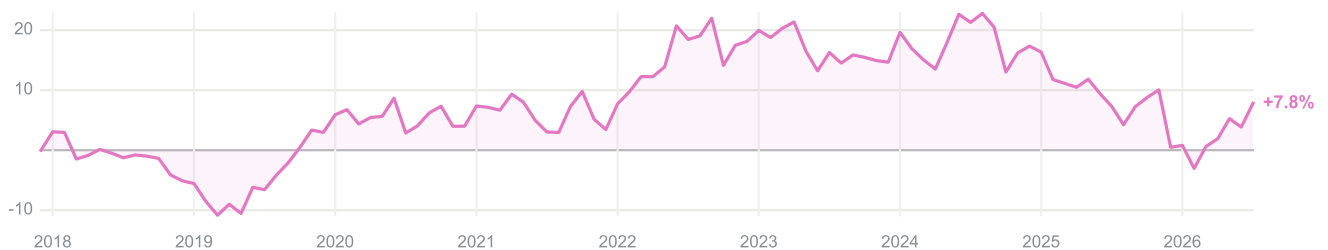
CAGR +0.7%



185 ● Cash Conversion Cycle

Earnings Quality

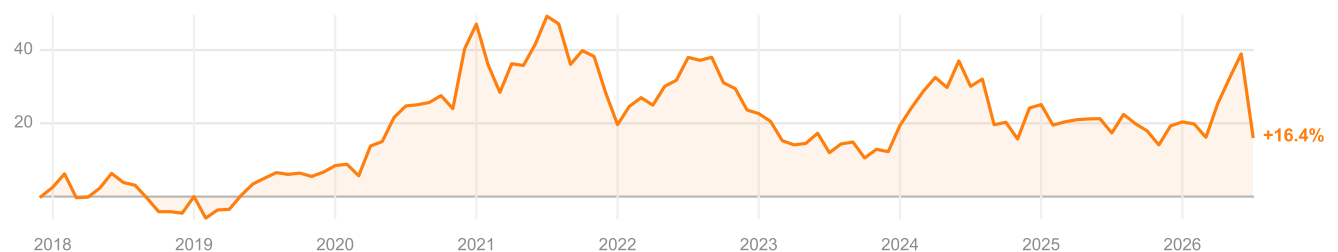
CAGR +0.7%



186 ● Volatility adjusted 12M return

Price Momentum

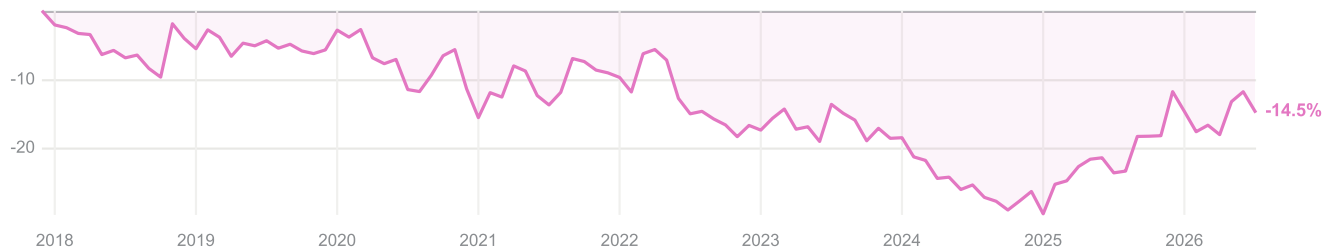
CAGR +0.6%



187 ● 1Y Chg in Sales to Earnings

Earnings Quality

CAGR +0.6%



188 ● Sustainable Growth Rate

Historical Growth

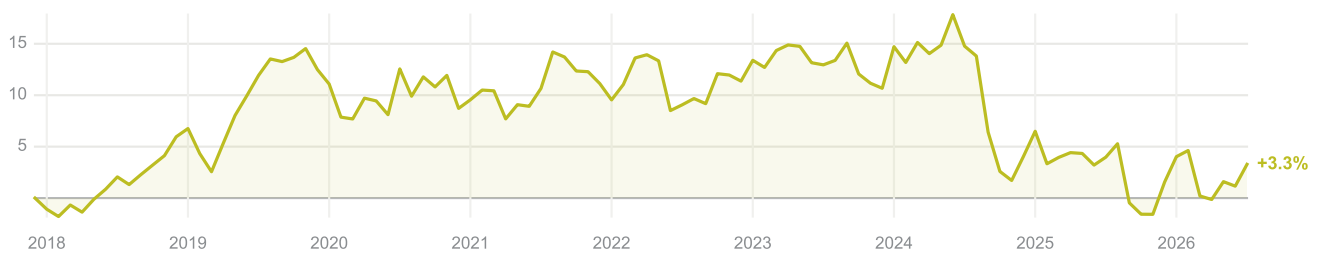
CAGR +0.6%



189 ● Year over Year Change of Total Debt

Capital Efficiency

CAGR +0.5%



190 ● Normalized ROE

Capital Efficiency

CAGR +0.5%



191 ● Capital Expenditure to Total Assets

Capital Efficiency

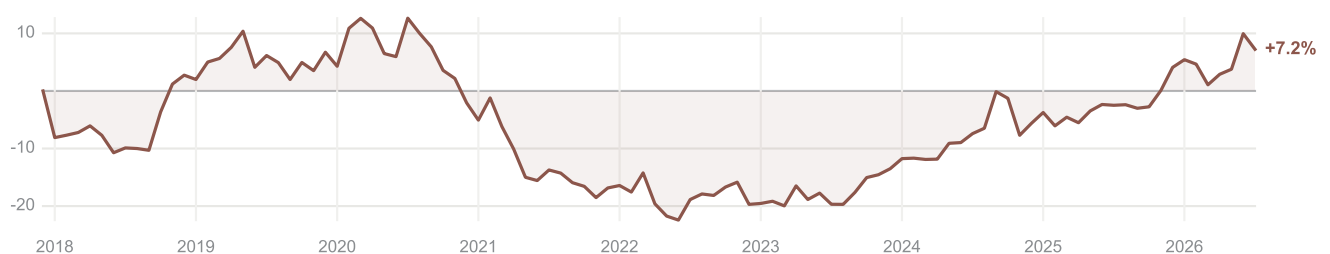
CAGR +0.5%

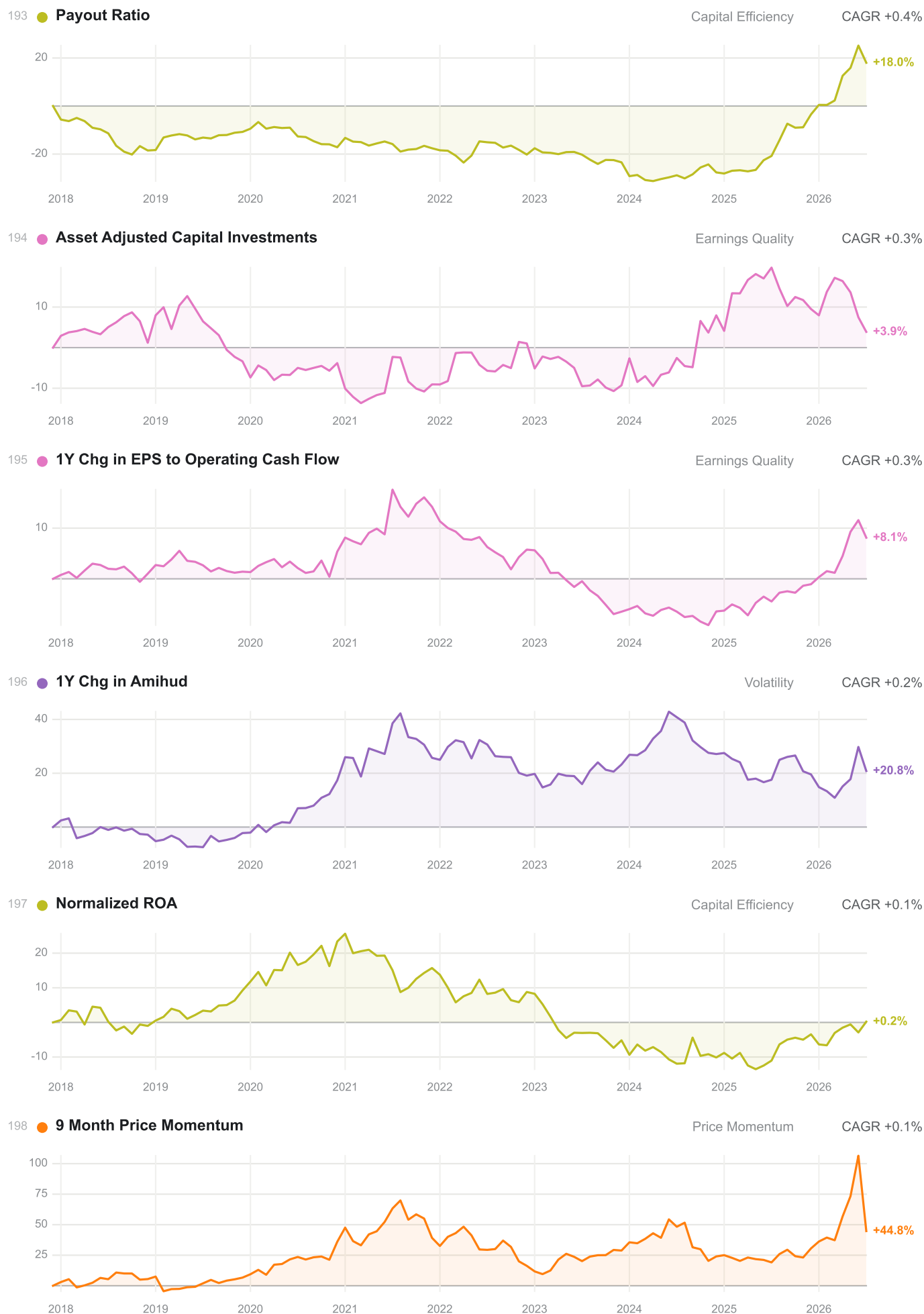


192 ● Trailing 12-Month Overhead/Trailing 12-Month Sales

Historical Growth

CAGR +0.4%

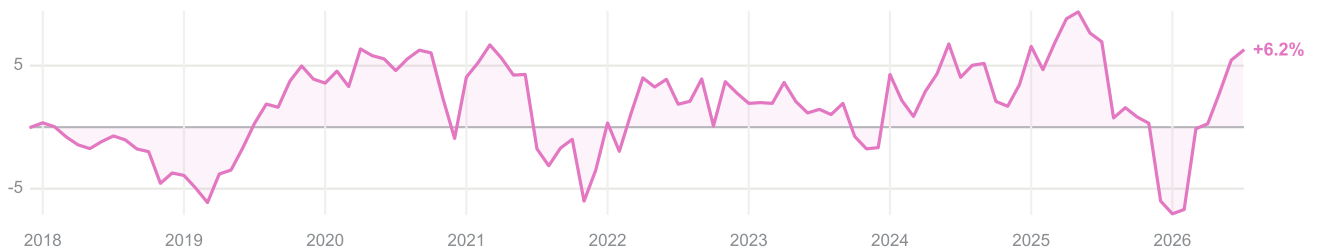




199 ● **Balance Sheet Accruals**

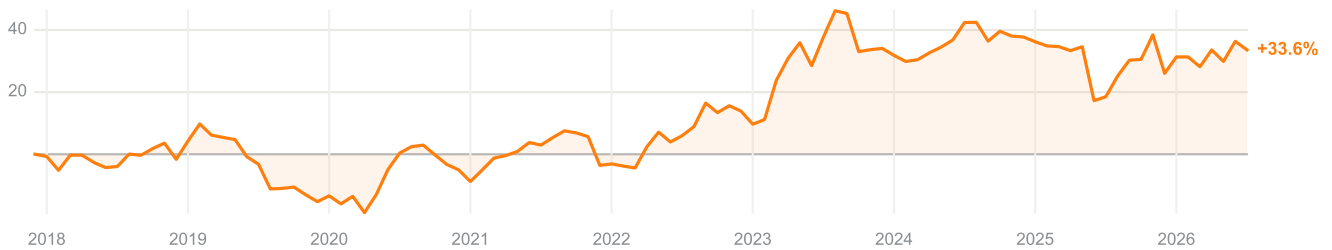
Earnings Quality

CAGR +0.1%

200 ● **6M Chg in 12M CAPM Alpha**

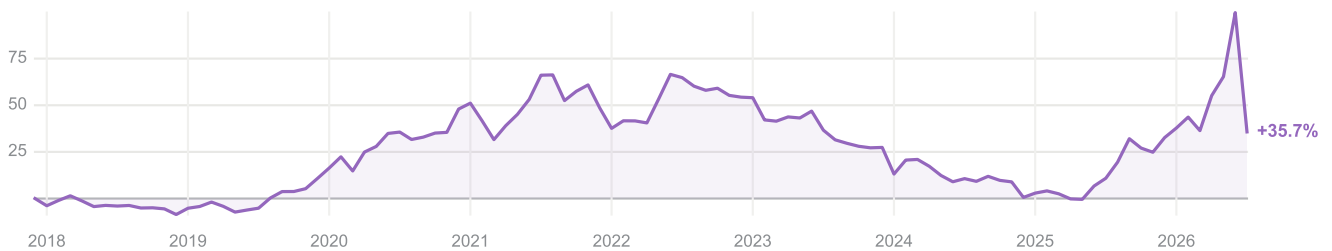
Price Momentum

CAGR +0.0%

201 ● **1M Realized Price Volatility**

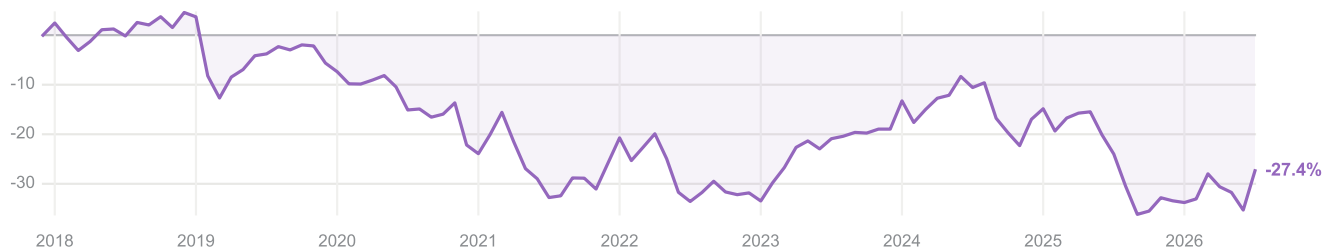
Volatility

CAGR +0.0%

202 ● **130 Day Minimum Return**

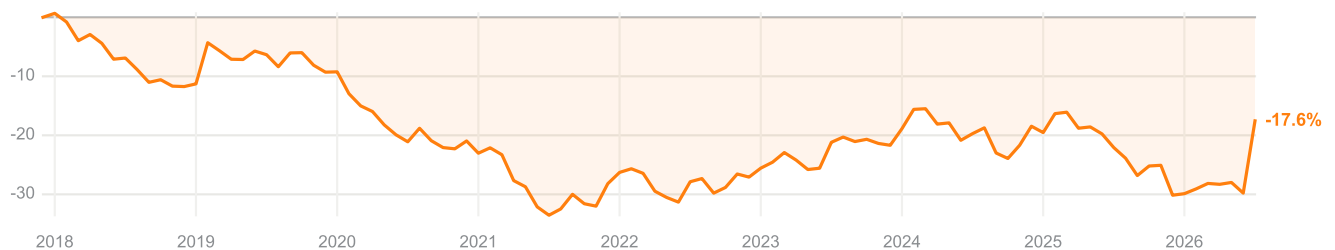
Volatility

CAGR +0.0%

203 ● **5 Day Price Reversal**

Price Momentum

CAGR -0.1%

204 ● **Analyst Dispersion for FY2 EPS**

Analyst Expectations

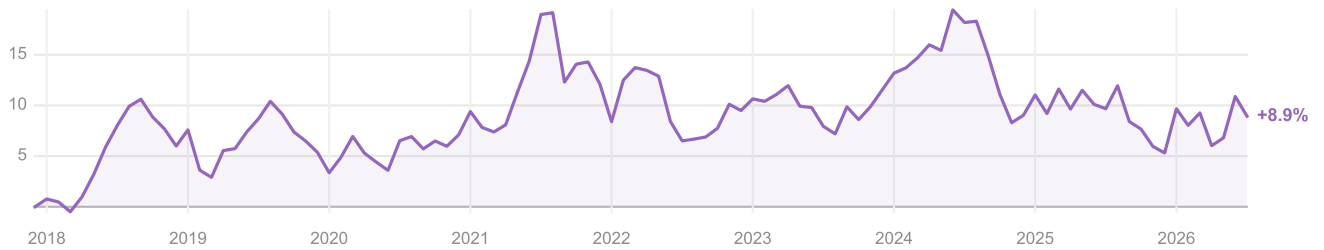
CAGR -0.2%



205 ● 1Y Chg in Debt to Assets

Volatility

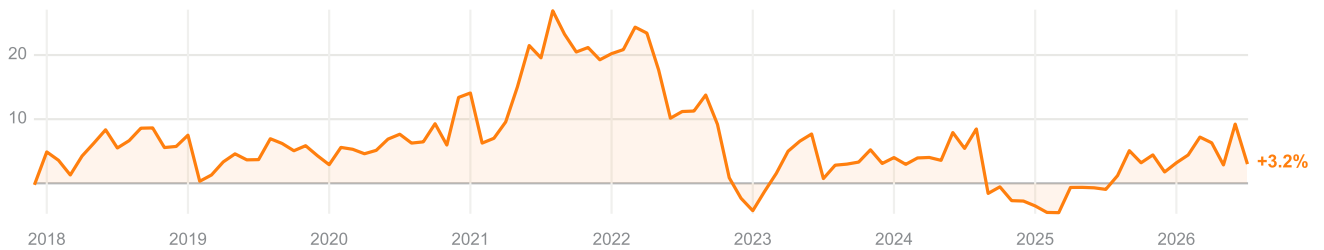
CAGR -0.2%



206 ● 26W Relative Price Strength

Price Momentum

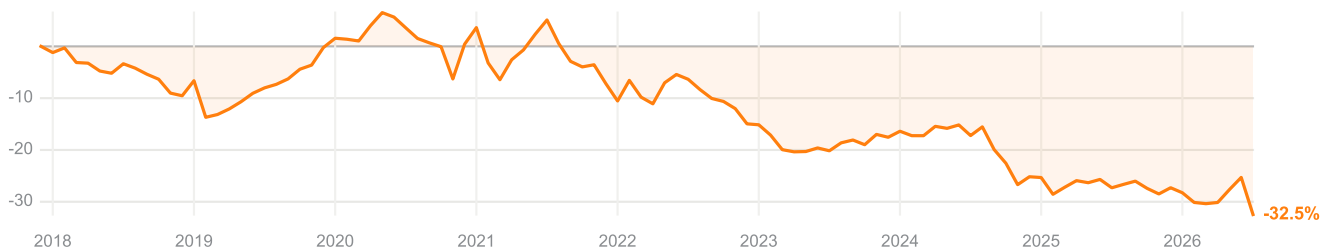
CAGR -0.2%



207 ● Change in Number of Holders

Price Momentum

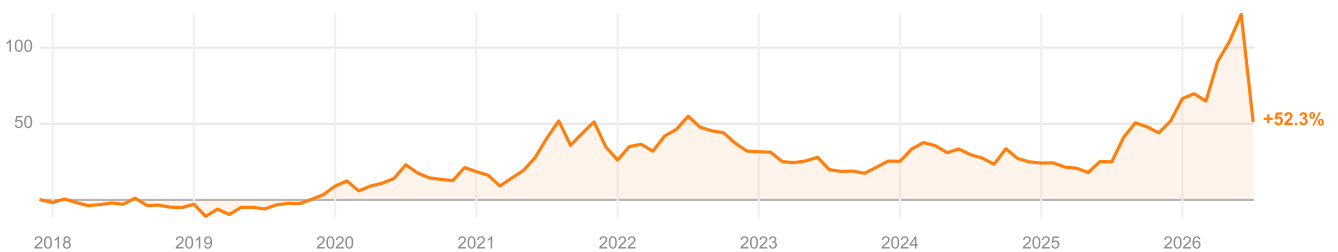
CAGR -0.4%



208 ● 52-Week Volume Price Trend with 20-Day Lag

Price Momentum

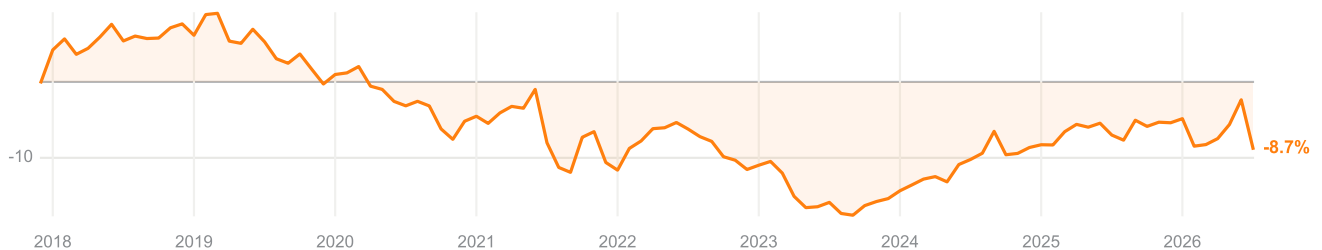
CAGR -0.4%



209 ● 6M Chg in 36M CAPM Alpha

Price Momentum

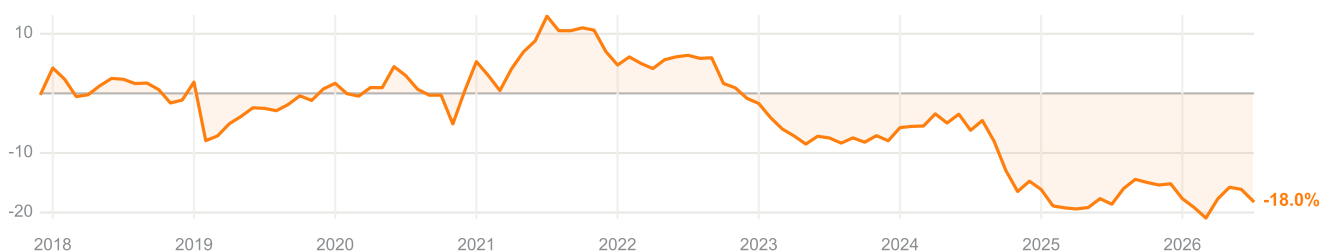
CAGR -0.6%



210 ● Stability of Change in Number of Holders

Price Momentum

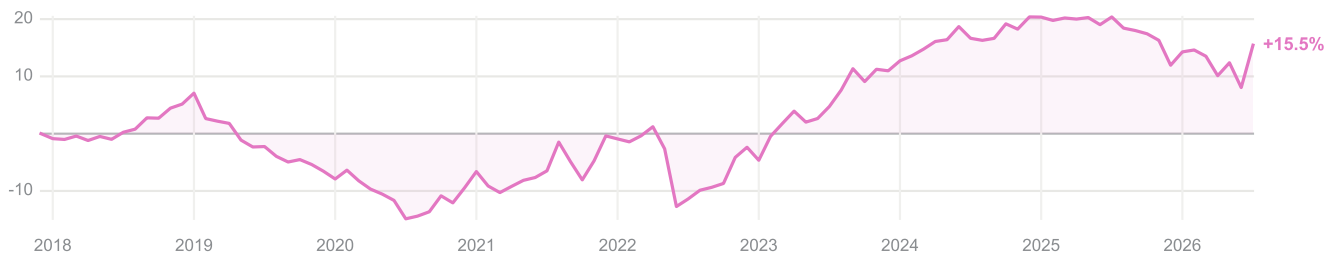
CAGR -0.7%



211 ● Sales to Gross Profit Margin Growth

Earnings Quality

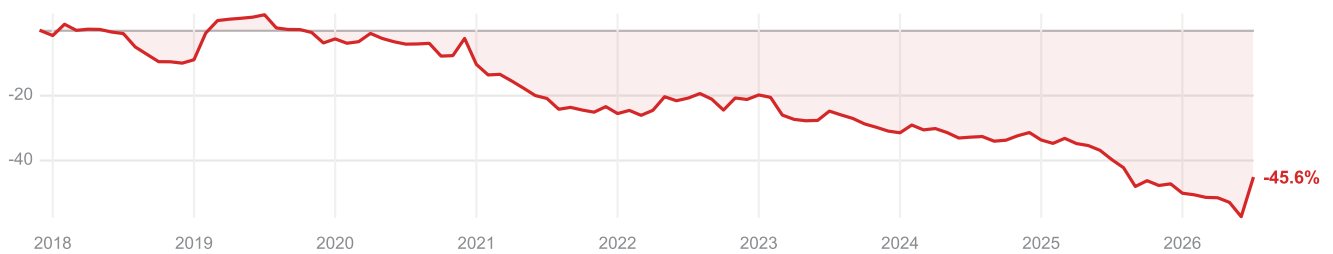
CAGR -0.7%



212 ● 4-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation

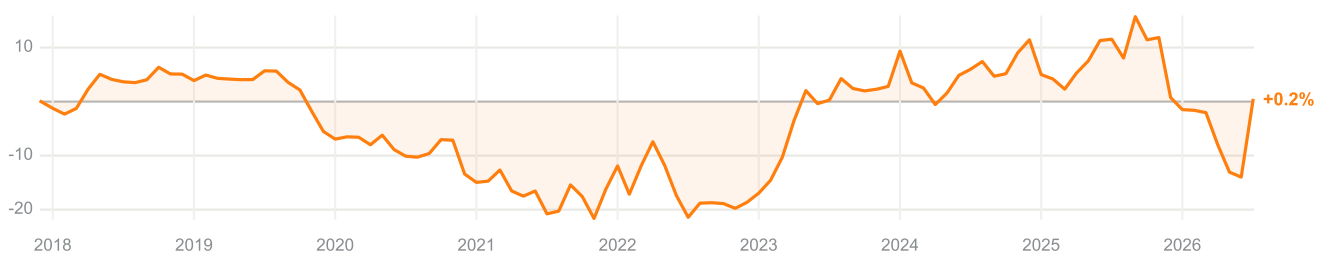
CAGR -0.7%



213 ● Institution Ownership Turnover

Price Momentum

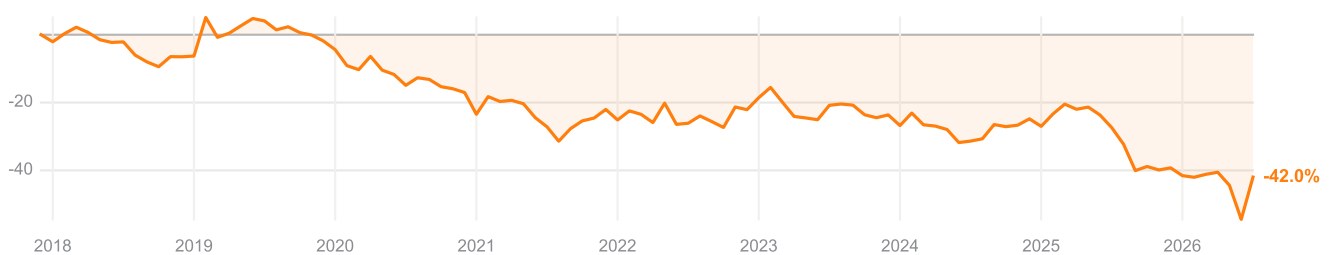
CAGR -0.8%



214 ● 1M Price Reversal

Price Momentum

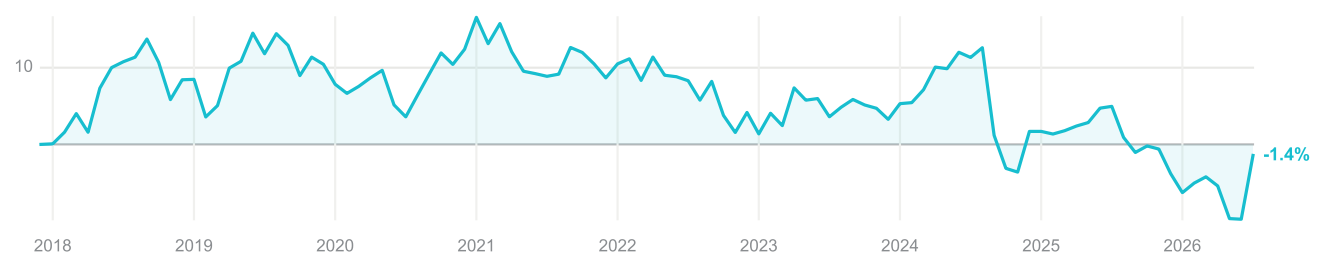
CAGR -0.8%



215 ● Revenue Estimate Dispersion

Analyst Expectations

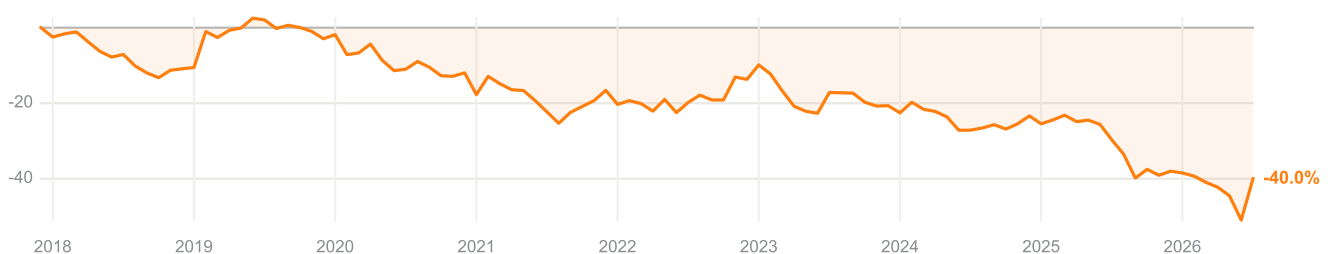
CAGR -1.0%



216 ● 14-Day Relative Strength Index

Price Momentum

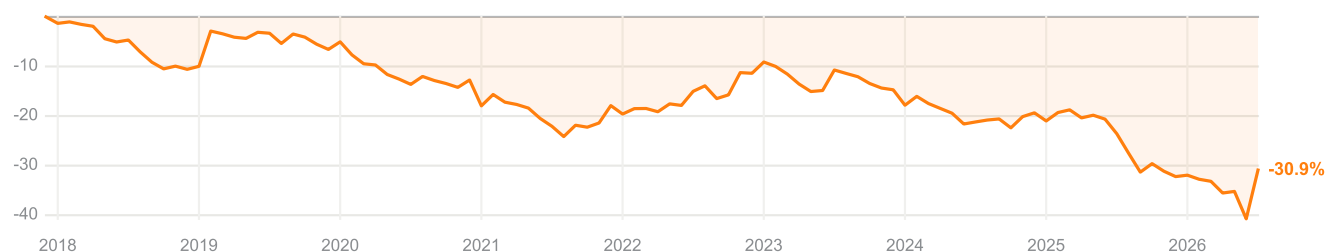
CAGR -1.1%



217 ● 1M Price High - 1M Price Low

Price Momentum

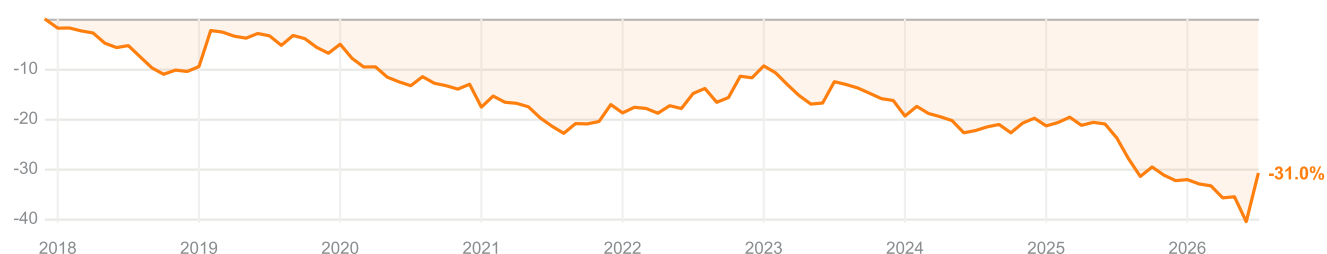
CAGR -1.1%



218 ● 20-Day Lane's Stochastic Indicator

Price Momentum

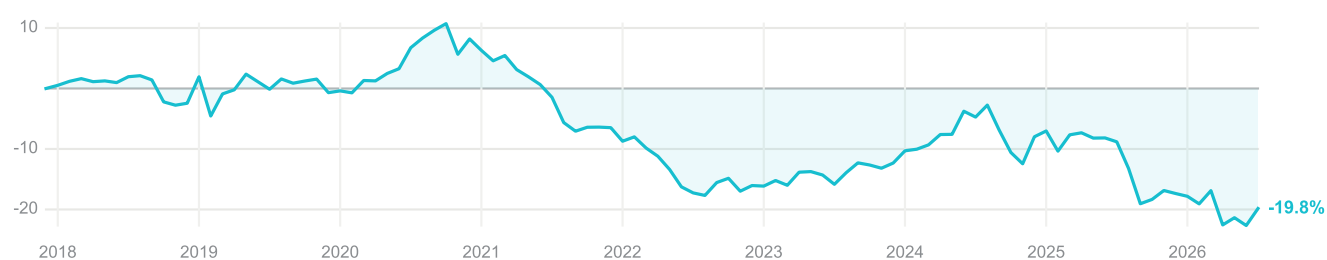
CAGR -1.2%



219 ● EBIT Estimate Dispersion

Analyst Expectations

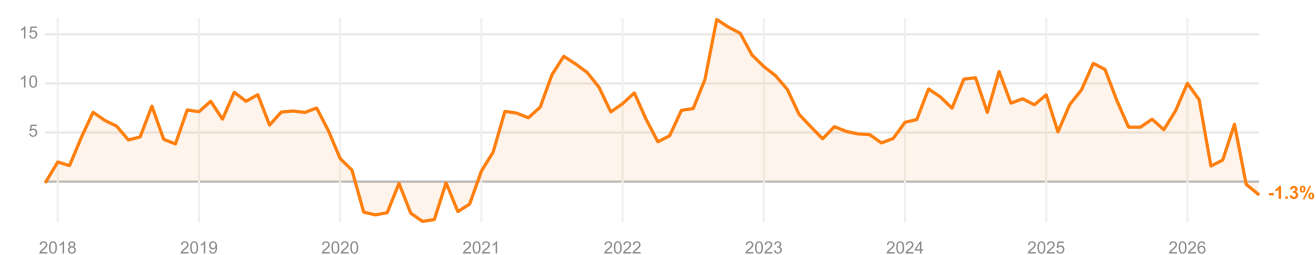
CAGR -1.7%



220 ● 6M Chg in 18M CAPM Alpha

Price Momentum

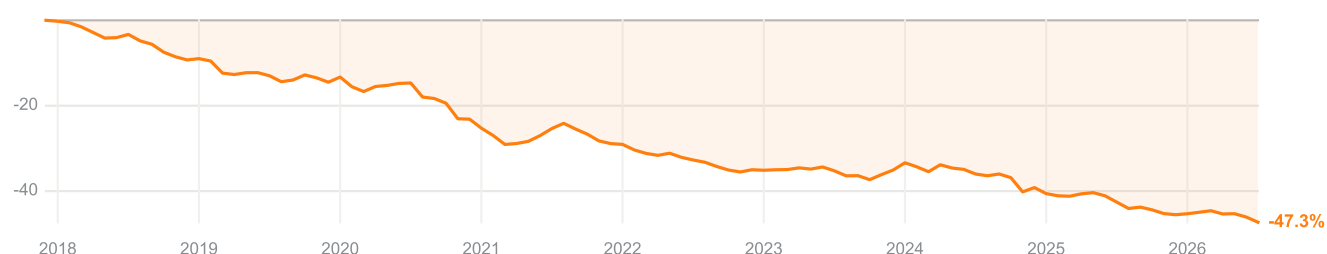
CAGR -2.0%



221 ● Change of Ownership Level - All Managers

Price Momentum

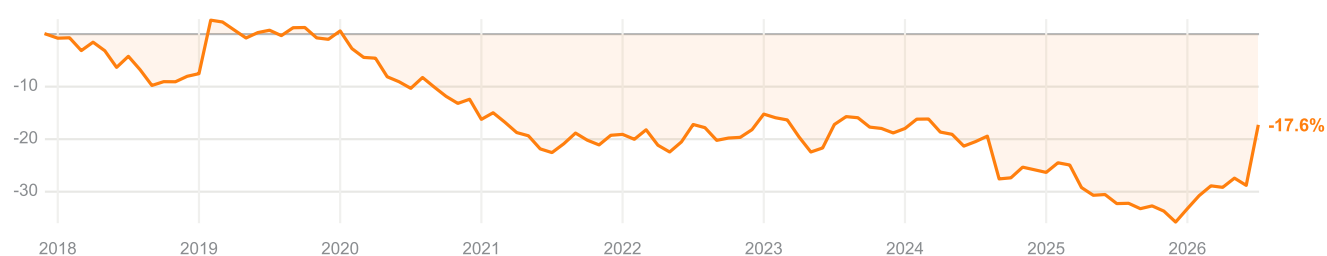
CAGR -2.2%



222 ● 5-Day Money Flow/Volume

Price Momentum

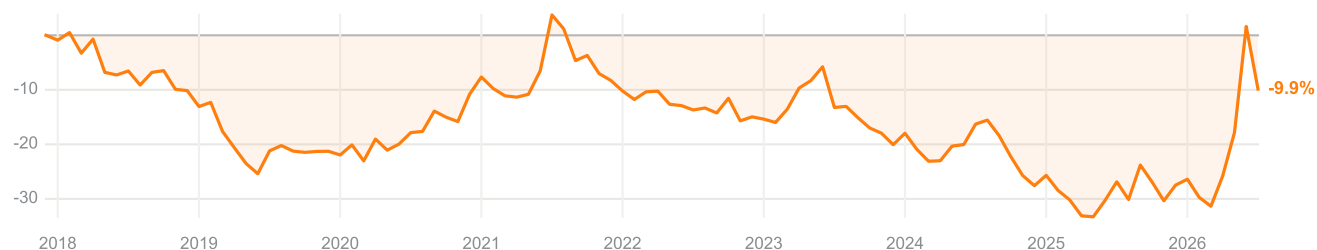
CAGR -2.6%



223 ● 5 Day Volume Signal

Price Momentum

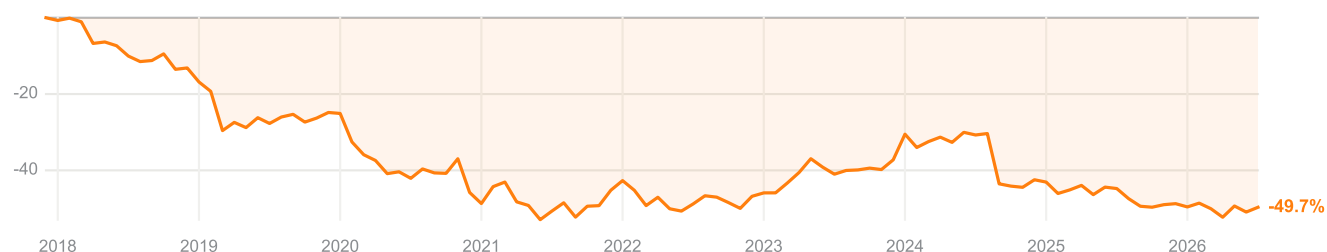
CAGR -4.5%



224 ● 20 Day Coefficient of Variation of Volume to Price

Price Momentum

CAGR -5.6%



225 ● 30 Day Coefficient of Variation of Volume to Price

Price Momentum

CAGR -6.0%

